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Investigation of short-term effects of forage radish on water and solute transport through a tracer test and inverse modeling

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Abstract

The benefits of using cover crops for improving soil and water quality are well known. Less clear is whether cover crops, especially those forming a taproot system, can favor solute transport down to the groundwater by modifying soil hydraulic properties and solute dynamics. In this study, we employed 12 lysimeters to conduct a comparative analysis between a taproot cover crop, specifically forage radish (FR), and bare soil (BS), under three water table management conditions. Our objective was to evaluate whether the enhancement of root-derived macroporosity could have modified water and solute dynamics, and offset the benefits provided by FR that is commonly used to mitigate solute leaching. A tracer solution of bromide (Br^-) was added to lysimeters, and solute flux concentrations were determined at different depths during a 25-day test. Soil moisture and pressure heads were monitored. Water and solute transport parameters were estimated by inverse modeling using HYDRUS-1D. A complementary laboratory experiment was performed to quantify the effect of FR root apparatus on the macropore structure by using noninvasive X-ray microtomography (μCT). Results showed that the growth of FR within the lysimeters induced alterations in water and solute dynamics compared with BS. This is primarily attributed to its proficiency as solute scavenger, with an uptake capacity of up to 47% of the total injected tracer. Our comparative analysis instead revealed subtle differences in soil structure and hydraulic properties brought about by the presence of FR. Major changes were observed for the saturated hydraulic conductivity (K_s), which increased from an average of 8.4–49.8 cm day^{-1} within the 20–45 cm layer in BS and FR, respectively. Additionally, there was a difference in immobile water content (θ_{im}), with the values in FR averaging 21% lower than those in BS. These modifications can be attributed to the formation of fissures and channels, primarily concentrated in the proximity of taproot development, without extending into deep preferential flow pathways. These structural changes were supported by the nondestructive μCT analyses. Upon

Abbreviations: μCT , X-ray microtomography; ADE, advection–dispersion equation; AIC, Akaike's information criterion; BS, bare soil; FD, free drainage; FR, forage radish; MIM, mobile–immobile equation; VWC, volumetric soil water content; WT120, water table level at 120-cm depth; WT60, water table level at 60-cm depth.

aggregating the effects observed, solute movement to groundwater was not affected by FR compared to BS conditions.

1 | INTRODUCTION

In recent decades, there has been renewed interest in the use of cover crops to improve soil functions (Adetunji et al., 2020; Blanco-Canqui et al., 2011; Scavo et al., 2022). Yunusa and Newton (2003) presented the concept of “primer plants,” namely, cover crops capable of conditioning the soil that is used for cash crops and conserving environmental resources. Although the use of cover crops is not a new concept, it is still debated to what extent cover crops can improve soil properties and soil-related ecosystem services. Particular attention has been put forth to cover crop benefits to mitigate greenhouse gas emissions, nutrients leaching, and soil degradation processes due to, for example, water erosion and loss of organic matter (Blanco-Canqui et al., 2015; Camarotto et al., 2018; Thorup-Kristensen et al., 2003). Recent studies have also focused on the effects of cover crops to improve soil hydraulic properties, leading to better regulation of water and nutrient cycles (Çerçioğlu et al., 2019; Patra et al., 2019).

Of particular interest are cover crops used in conservation agriculture, as the implementation of no-tillage practices may lead to some topsoil compaction (Chen et al., 2014), reduce air and water exchange, and in turn reduce yields (Ball et al., 1994; Carter, 1991). In this case, taproot cover crops could help prevent soil compaction by improving the soil-pore arrangement, water infiltration, and water holding capacity, especially in soils prone to degrade their naturally weak structure (Munkholm et al., 2003), although some excessive compaction may cause taproots to grow aboveground and struggle to penetrate and fracture soil, leaving the ability to grow into deep soil layers dependent on the smaller (non-fleshy) portion of the root (Cofer, 2019).

The ability of a cover crop to modify the soil structure depends on the characteristics of its growth pattern (Ramírez-García et al., 2015), especially the root architecture and the quantity and characteristics of the root biomass (Chen & Weil, 2010). It has been suggested that the root diameter is the main trait to effectively prime the soil pore space. Coarse roots allow more effective movement of the soil particles and have a lower tendency to curl under mechanical stress conditions (Clark et al., 2003). Taproot cover crops have shown greater biological drilling potential than fibrous root plants (Cresswell & Kirkegaard, 1995; Marshall et al., 2016). Taproots can create large bio-channels and fissures that increase the flow of water and air into the soil (Guo & Lin, 2018), which in turn provide benefits to the soil system by facilitating the root

access of the main crop to the deeper layers (Chen & Weil, 2011).

Forage radish (FR; *Raphanus sativus* L. var. *longipinnatus* L. H. Bailey) is a taproot plant belonging to the Brassicaceae family that has drawn the attention of farmers in recent years for its alleged capacity to provide adequate soil cover, scavenge nutrients, suppress weeds (Gruver et al., 2014), and prevent soil compaction. The latter has led some farmers to name FR as “tillage radish” or “radish ripper” (Weil et al., 2009). FR grows best in cool, moist growing conditions and is not tolerant of shade, stagnant water, or severely nitrogen-deficient soils (Weil et al., 2009). Planting times should allow at least 2 months of plant growth before winterkill, which occurs below -7°C (Sundermeier, 2008). Typical seeding rates for a pure stand of FR are $7\text{--}10\text{ kg ha}^{-1}$ (Weil et al., 2009). Under favorable growing conditions, the roots of FR can extend down to 1-m depth in only 60 days, with their thickened storage portion extending down to 30 cm (Kristensen & Thorup-Kristensen, 2014). After FR winterkill, the large roots desiccate, open channels are created up to the soil surface, and infiltration, drainage, and soil warming are improved (Gruver et al., 2014). Moreover, the large biopores left by FR can reduce the resistance of soil for root growth of subsequent cash crop roots (Williams & Weil, 2004). This effect is emphasized in no-tillage systems because undisturbed soil structure is mostly maintained over time. Chen and Weil (2010) detected a greater ability of FR to penetrate compacted soils than other cover crops, such as rye and rapeseed. Abdollahi and Munkholm (2014) proved that FR created continuous pores with a diameter $>30\text{ }\mu\text{m}$ and, in turn, improved water and gas transport, while Linsler et al. (2016) found a greater proportion of large and medium macroaggregates after growing FR compared to fallow. However, there could be drawbacks from creating such large pores, such as improper seed establishment of the subsequent main crop, or the risk of preferential flows that could bypass nutrients and pesticide filtration by soil (Beven & Germann, 2013) and facilitate leaching, especially under shallow groundwater conditions (Jarvis, 2007; Mencaroni et al., 2023; Shipitalo et al., 2000). This is emphasized in no-tillage systems, where extensive use of pesticides for weed control and cover crop devitalization must offset unforeseen tillage operations. Nevertheless, it is worth noting that other studies reported contrasting findings (e.g., Piccoli et al., 2022). They observed minimal alterations, for example, in soil hydraulic parameters at varying degrees of root decay after cover crop devitalization, suggesting that the different occupation of the porosity by the roots can modify

water and solute infiltration dynamics. All these considerations remain important even during the plant's growth phase, where a compensatory balance is established between solute leaching and root uptake.

In this context, at an experimental lysimeter site, we aimed to estimate the effects of FR on water and solute transport dynamics during the growing season as affected by structural changes and understand the vulnerability of agroecosystems under different water table management conditions. A complementary laboratory experiment was also performed to quantify by X-ray computed microtomography the effect of soil root apparatus on the macropore architecture. Our starting hypothesis was that the benefits of using FR to mitigate nutrient leaching can be offset by enhanced root-derived macroporosity, which fosters preferential solute movement down to the groundwater.

2 | MATERIALS AND METHODS

2.1 | Site description

A leaching experiment was conducted at the Experimental Farm "L. Toniolo" of the University of Padova in Legnaro, NE Italy (45°21'N, 11°57'E, 8 m a.s.l.) in 2017. The local climate (reference period 1964–2015) is subhumid with an annual average temperature and rainfall of 12°C and 800–850 mm, respectively (Dal Ferro et al., 2017; Giardini, 2004). Annual rainfall is mostly concentrated during the autumn and spring months, and the reference evapotranspiration (ET_0) is 800 mm. The study site was originally set up in 1984, consisting of twenty 1 m × 1 m × 1.5 m (length × width × depth) drainable lysimeters. The bottom of each lysimeter is funnel-shaped and connected via an underground drainpipe (1‰ slope) to an external tube with a 3-cm diameter hose to avoid the effect of capillary action, equipped with a valve to regulate both the water table level and leaching discharge (Figure 1). The external tube is equipped with a water level sensor that can be used to manually regulate the water level, as in Longo et al. (2021).

Lysimeters were filled until 140-cm depth with soil excavated from the adjacent experimental farm by preserving the original soil horizons (Table 1). To facilitate water drainage and prevent soil washout, the deepest 10-cm-thick layer was filled with gravel (30- to 50-mm diameter). The soil was silty loam classified as Fluvi-Calcaric Cambisol (IUSS Working Group WRB, 2015).

2.2 | Experimental design

A total of six treatments were tested in a factorial combination of two soil cover conditions, that is, FR and bare soil (BS),

Core Ideas

1. Taproot served as solute scavenger, offsetting the risk of leaching at –60 cm, below its fleshy portion.
2. Taproot-induced soil hydraulic changes were mainly observed in the 0–45 cm layer.
3. Forage radish caused an increase in the macroporosity where the taproot mainly developed.

with three groundwater managements, that is, a water table level at 60-cm (WT60) and 120-cm (WT120) depths, and free drainage (FD). The water table was regulated manually on a daily basis throughout the entire duration of the test. Treatments were replicated twice (the two replications are hereafter distinguished by the subscripts "a" and "b") following a completely randomized design. The number of replicates was limited by the high cost of the equipped lysimeters, as already reported in Longo et al. (2023). FR was sown on September 28, 2016, with a density of 40 plants m⁻², and fertilized with ammonium nitrate (NH₄NO₃) (30 kg N ha⁻¹ year⁻¹) just before sowing. In contrast, BS conditions were maintained throughout the test (no sowing) manually eliminating any germinating weeds. Note that before starting the experiment, all lysimeters were managed according to the same tillage operations (Longo et al., 2021). Two lysimeters, FR-FD_a and BS-FD_a, showed failures in soil hydraulic parameter acquisition during the trial and were not further considered, thus bringing the total number of lysimeters investigated regarding soil properties to 10 (five FR and five BS).

2.3 | Automated monitoring and pore-water sampling systems

An automated system was set up to monitor soil water and temperature dynamics at depths of 15, 30, and 60 cm (Figure 1) (Mencaroni et al., 2021). The system consisted of 36 CS635 TDR (Time Domain Reflectometry) probes (Campbell Scientific Inc.) connected to a TDR100 Time Domain Reflectometer (Campbell Scientific Inc.) to measure the volumetric water content; 36 T4e electric tensiometers (UMS GmbH) to measure the soil matric potential; and 24 temperature thermistor probes (Campbell Scientific Inc.), installed only at 15- and 30-cm depth, to measure soil temperature. All the probes were connected to a CR-10X datalogger (Campbell Scientific Inc.) through a series of multiplexers. The system was also equipped with 36 SPE20 polyethylene/nylon suction cups (Meter Group) to collect the pore water.

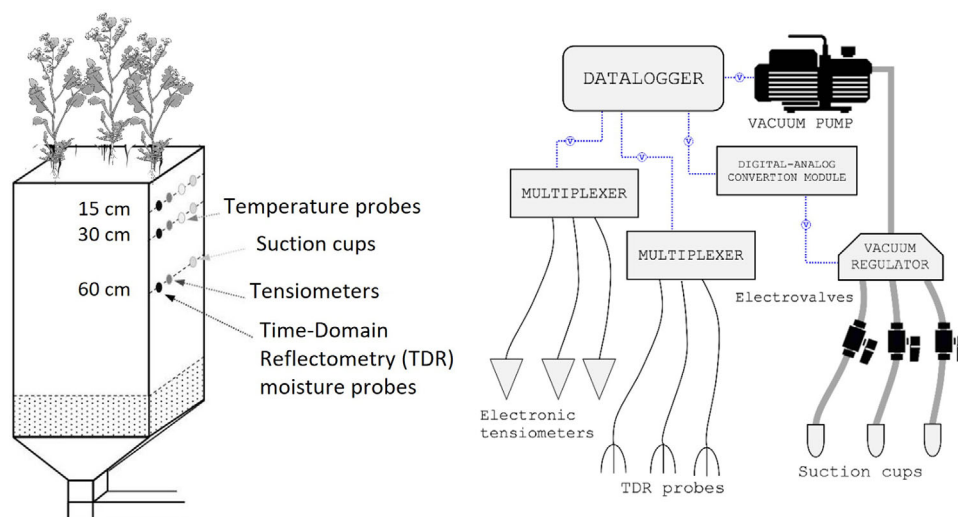


FIGURE 1 Sketch of a lysimeter equipped with sensor probes and pore water samplers (left), and scheme of the automated monitoring and pore-water sampling (right).

TABLE 1 Mean soil physical and chemical properties $\pm$ standard error at different depths at the study site.

Soil properties	Soil depth			
	0–20 cm	20–40 cm	55–70 cm	95–110 cm
Sand (%)	27 $\pm$ 0.9	25 $\pm$ 1.3	20 $\pm$ 2.4	18 $\pm$ 0.6
Silt (%)	56 $\pm$ 0.6	57 $\pm$ 0.7	60 $\pm$ 2.1	61 $\pm$ 1.2
Clay (%)	17 $\pm$ 0.6	18 $\pm$ 0.7	20 $\pm$ 0.4	21 $\pm$ 0.5
pH (H ₂ O)	7.8 $\pm$ 0.1	8.2 $\pm$ 0.1	8.3 $\pm$ 0.0	8.3 $\pm$ 0.0
EC (μ S cm ⁻¹)	267 $\pm$ 13.8	192 $\pm$ 10.6	194 $\pm$ 6.5	190 $\pm$ 5.0
Bulk density (g cm ⁻³)	1.27 $\pm$ 0.08	1.35 $\pm$ 0.07	1.44 $\pm$ 0.06	1.45 $\pm$ 0.05
SOC (%)	0.92 $\pm$ 0.05	0.71 $\pm$ 0.04	0.59 $\pm$ 0.03	0.61 $\pm$ 0.02
N (%)	0.12 $\pm$ 0.01	0.10 $\pm$ 0.01	0.09 $\pm$ 0.01	0.09 $\pm$ 0.01
P (mg kg ⁻¹)	44.3 $\pm$ 12.5	21.8 $\pm$ 3.4	11.6 $\pm$ 3.6	17.5 $\pm$ 3.2
CEC (meq 100 g ⁻¹)	14.2 $\pm$ 0.3	14.7 $\pm$ 0.2	17.2 $\pm$ 0.7	17.2 $\pm$ 0.5
Total porosity (m ³ m ⁻³)	0.46 $\pm$ 0.01	0.44 $\pm$ 0.01	0.45 $\pm$ 0.00	0.42 $\pm$ 0.01
SWC at –33 kPa (m ³ m ⁻³)	0.28 $\pm$ 0.01	0.29 $\pm$ 0.01	0.32 $\pm$ 0.01	0.30 $\pm$ 0.01
SWC at –1500 kPa (m ³ m ⁻³)	0.15 $\pm$ 0.01	0.16 $\pm$ 0.01	0.18 $\pm$ 0.01	0.15 $\pm$ 0.00

Abbreviations: CEC, cation exchange capacity; EC, electrical conductivity (1:2 soil-water ratio); N, total nitrogen content; P, extractable phosphorus content by Olsen method; SOC, soil organic carbon content; SWC, soil water content.

The automatic system allowed to maintain suction in equilibrium with the surrounding soil water tension (Morari, 2006) by integrating soil matric potential measurements, acquired by the tensiometers, with the pore water sampling system. To regulate the vacuum on pore water samplers based on matric potential, a system composed of a 5-V digital-analog conversion module (mod. A04A; Campbell Scientific Inc.) connected to an electronic vacuum regulator incorporating a pressure relief valve (ITV2091, SMC Corporation) was employed, which adapted the digital pressure to an analog electric signal and continuously regulated the vacuum on a vacuum pump. The imposed negative pressure, applied to

the soil-water sampler from –2 to –70 kPa, was maintained close to the soil water tension readings. Therefore, the system overcame frequent vacuum fluctuations of traditional systems that controlled the suction application duration or were subjected to vacuum pressure leakage in vacuum tanks (Farsad et al., 2012).

2.4 | Tracer test

A total of 10 L of saline tracer solution of KBr (Levy & Chambers, 1987), containing 667 mg L⁻¹ of Br⁻ concentration

(6667 mg of Br^- in total), was added to each lysimeter, simulating a rainfall event of 10 mm on March 27, 2017. Water sampling started the day before rainfall simulation and ended on April 21, 2017. Water samples were collected every day for the first 2 weeks and every other day for the following weeks. Once collected, water samples were stored at -20°C before analysis, which consisted of Br^- concentration detection with an Ion-Meter (pH & ION meter GLP 22; Crison). Prior to the tracer test, the background Br^- concentration of the soil solution was assessed at different depths and in the leaching water. During the tracer test, the plants of FR were in full vegetative growth.

2.5 | Plant sampling and analysis

The aboveground crop biomass in all the FR lysimeters (six in total) was harvested on April 26, 2017. Additionally, five plants per lysimeter were randomly selected, and both the swollen taproot and the roots attached to it were manually harvested to determine their biomass. This approach aimed to minimize soil structure alterations, thus avoiding any compromise to subsequent experiments (Longo et al., 2021). The root samples underwent thorough washing to eliminate any soil residues, and, following that, all biomass samples were dried for 72 h at 65°C in a forced draft oven to determine their dry weight. Samples of dry biomass were subjected to microwave closed vessel digestion (Ethos 1600 Milestone S.r.l. Sorisole) and then analyzed with ICP-OES, Arcos II MV (Spectro A. I. GmbH) to measure Br^- concentration. Aboveground and taproot dry biomass, and Br^- concentration and uptake were tested according to one-way analysis of variance including the water table level as treatment factor at $p < 0.05$.

2.6 | Numerical model of flow and transport

Effects of FR on hydraulic and solute transport parameters were estimated using the HYDRUS-1D model (Šimůnek et al., 2008) by applying the model inversion method. Inverse modeling is a process that consists of searching for the best set of parameters, varying them iteratively, and comparing the variables provided by the model with corresponding values measured under field conditions (Šimůnek et al., 1999). This is one of the most popular methods currently used due to its ability to provide good estimates of soil parameters from relatively simple data (Le Bourgeois et al., 2016; Ritter et al., 2003).

The HYDRUS-1D software (Šimůnek et al., 2008) is a one-dimensional (1D) hydraulic simulation model of water flow and solute transport in incompressible, variably saturated porous media in steady or transient regime (Kanzari et al., 2018). HYDRUS-1D uses a Galerkin finite-element method

to solve the Richards equation augmented with a sink term to represent the root water uptake:

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z} \left[K(h) \left(\frac{\partial h}{\partial z} + 1 \right) \right] - S(h) \quad (1)$$

where θ is the volumetric water content [$\text{L}^3 \text{L}^{-3}$], h is the soil water pressure head [L], z is the vertical coordinate [L], t is the time [T], K is the unsaturated hydraulic conductivity function [L T^{-1}], and $S(h)$ is the sink term for the root water uptake [$\text{L}^3 \text{L}^{-3} \text{T}^{-1}$] (Šimůnek & van Genuchten, 2008). The volumetric water flux density q [L T^{-1}] computed by the generalized Darcy's law reads as:

$$q(h) = K(h) \left(\frac{\partial h}{\partial z} + 1 \right) \quad (2)$$

The classical van Genuchten–Mualem model (van Genuchten, 1980) was used to represent the water retention curve [$S_e(h)$] and the unsaturated hydraulic conductivity function [$K(h)$]:

$$S_e = \frac{\theta - \theta_r}{\theta_s - \theta_r} \quad (3)$$

$$S_e(h) = \left[\frac{1}{1 + (\alpha h)^n} \right]^m \quad (4)$$

$$K(h) = K_s S_e^l \left[1 - \left(1 - S_e^{\frac{1}{m}} \right)^m \right]^2 \quad (5)$$

where S_e is the relative saturation ($0 \leq S_e \leq 1$) (-); θ_s and θ_r are the saturated and residual water content, respectively [$\text{L}^3 \text{L}^{-3}$]; K_s is the saturated soil hydraulic conductivity [L T^{-1}]; l is the pore connectivity parameter (-); and α [L^{-1}], n (-), and m ($m = 1 - 1/n$) are empirical parameters describing the shape of the retention curve. The ratio $1/\alpha$ is assimilated to the air-entry suction value (or bubbling pressure), while n is a pore-size distribution index that determines the curve slopes. Six parameters were thus necessary to define the hydraulic properties of each lysimeter: θ_r , θ_s , α , n , K_s , and l .

For the lysimeters cultivated with FR, the root water uptake was considered. The sink term, S in (1), represents the volume of water removed per unit time from a unit volume of soil due to plant water uptake. Feddes et al. (1976) defined S as a function of h :

$$S(h) = \alpha_r(h) S_p \quad (6)$$

$$S_p(z) = \beta_r(z) T_p \quad (7)$$

where the root water stress response function $\alpha_r(h)$ is a prescribed dimensionless function of the soil water pressure head ($0 \leq \alpha_r \leq 1$), S_p is the potential water uptake rate [$\text{L}^3 \text{L}^{-3} \text{T}^{-1}$],

T_p is the transpiration rate [$L^3 L^{-2} T^{-1}$], and $\beta_r(z)$ [L^{-1}] is the normalized root density distribution such that $\beta_r(z) dz$ is the fraction of roots located between z and $z + dz$.

Solute transport under equilibrium and physical nonequilibrium state was simulated with the advection–dispersion equation (ADE) and the mobile–immobile (MIM) equation, respectively. The ADE is widely used to describe solute transport with Fickian dispersion behavior when all the water phase is contributing to the flow and is presented in Equation (8). It should be noted that adsorption is not taken into account since Br^- is considered to be a conservative tracer.

$$\frac{\partial \theta c}{\partial t} = \frac{\partial}{\partial z} \cdot \left(\theta D \frac{\partial c}{\partial z} \right) - \frac{\partial qc}{\partial z} - S(h)c \quad (8)$$

where c is the solute concentration [$M L^{-3}$] and D is the dispersion coefficient [$L^2 T^{-1}$] (Šimůnek & van Genuchten, 2008). The dispersion coefficient, accounting for both molecular diffusion and hydrodynamic dispersion, was defined as:

$$D = \xi(\theta) \cdot D_l^w + \lambda \frac{|q|}{\theta} \quad (9)$$

where $\xi(\theta)$ is the dimensionless tortuosity factor, D_l^w is the molecular diffusion coefficient in free water [$L^2 T^{-1}$], $|q|$ is the absolute value of the Darcian volumetric water flux density [$L T^{-1}$], and λ is the dispersivity [L], a parameter that expresses the mixing length of the solute in the soil. The tortuosity factor was calculated in HYDRUS as a function of the water content using the relationship of Millington and Quirk (1961):

$$\xi(\theta) = \frac{\theta^{7/3}}{\theta_s^2} \quad (10)$$

The ADE is widely used to simulate solute transport. However, it may fail to accurately describe the fate of solute in soil presenting a noticeable heterogeneity such as macropores or fractures associated with a fine matrix (Gao et al., 2010). When such a physical nonequilibrium occurs, the MIM equation (van Genuchten & Wierenga, 1976) should be applied as a modified version of the ADE. It assumes that the water flow and the associated solute transport (by advection and dispersion at the large scale) are restricted only to the macropores (mainly interaggregate pores and fractures) (Šimůnek & van Genuchten, 2008). Hence, the water inside micropores (mainly intra-aggregate pores) is stagnant, and solutes can only access this pore space by molecular diffusion at the small scale. In the MIM model, the water content θ is thus divided into two fractions, that is, the mobile phase (θ_{mo}), which plays a key role in solute leaching, and the immobile phase (θ_{im}):

$$\theta = \theta_{mo} + \theta_{im} \quad (11)$$

The MIM model uses two equations, one for the mobile domain (12) and the other for the immobile domain (13), without the sorption terms, as reported by Šimůnek and van Genuchten (2008):

$$\frac{\partial \theta_{mo} c_{mo}}{\partial t} = \frac{\partial}{\partial z} \cdot \left(\theta_{mo} D_{mo} \frac{\partial c_{mo}}{\partial z} \right) - \frac{\partial qc_{mo}}{\partial z} - \Gamma_s - S(h)c_{mo} \quad (12)$$

$$\frac{\partial \theta_{im} c_{im}}{\partial t} = \Gamma_s \quad (13)$$

where c_{mo} and c_{im} are concentrations of solutes in the mobile and immobile regions [$M L^{-3}$], respectively, D_{mo} is the dispersion coefficient in the mobile region [$L^2 T^{-1}$] computed with Equation (9) with θ_{mo} , and Γ_s is the rate of mass transfer between the two domains [$M L^{-3} T^{-1}$]. This solute transfer between the mobile and immobile domain is classically described by at first order as:

$$\Gamma_s = \alpha_{MIM} (c_{mo} - c_{im}) \quad (14)$$

where α_{MIM} [T^{-1}] is a mass transfer coefficient which describes the diffusive transport of solute in the geometric configuration of mobile and immobile domains at the small scale.

2.7 | Implementation of the inverse model

Soil hydraulic parameters were calibrated using an inverse solution procedure based on the Marquardt–Levenberg algorithm (Marquardt, 1963) implemented in HYDRUS-1D (REF) applied to Equations (1) and (3–7). It was performed considering multiple soil layers, that is, 0–20, 20–45, and 45–130 cm, for reproducing the heterogeneity of the soil profile as affected by soil and cover crop management. The model was initially run to parameterize a homogeneous soil profile, but results showed poor agreement with observed water dynamics. Calibrating many parameters simultaneously may enhance the likelihood of nonuniqueness and instability in the inverse solution (González et al., 2015). Thus, a stepwise layer-by-layer procedure starting from the top, applied to each lysimeter, was performed considering the parameters θ_s , α , n , and K_s , as described by Diederik et al. (2002). The parameters obtained by optimizing a homogeneous soil profile were used as starting points for the next layer-by-layer optimization. When the hydraulic parameters of all layers were estimated, some further fine tuning was done by reestimating the parameters of each layer until no further improvement was achieved. To reduce the number of unknowns, θ_r and l were fixed to values of $0.01 \text{ cm}^3 \text{ cm}^{-3}$ and 0.5, respectively, consistent

with previous data on the same soils (Mencaroni et al., 2021; Morari, 2006). The inversion was carried out on data from February 15 to April 25, 2017 (70 days in total), on a daily time scale. The daily average of soil volumetric water content and soil water pressure head at depths of 15, 30, and 60 cm were used as calibration data. The obtained results of soil hydraulic parameters were used to model solute transport dynamics. To note, at the beginning of the experiment, from February 15 to March 20, 2017, a failure in the multiplexer system prevented recording soil water pressure head data in a few lysimeters. In this period, the inversion was based on soil moisture data and on the data provided by the remaining tensiometers.

All weather parameters required for computation were recorded by a meteorological station located at the experimental site. The upper boundary conditions were set up as “atmospheric boundary conditions with surface layer,” given the absence of run-off in the lysimeters. Daily reference evapotranspiration (ET_0) was calculated according to the FAO-56 Penman–Monteith equation (Allen et al., 1998). The ET_0 was multiplied by a crop coefficient (K_c) to estimate the crop evapotranspiration (ET_c) of FR and BS (0.90 and 1.15, respectively) (Allen et al., 2005). Root water uptake was calculated by applying the plant water stress response function proposed by Feddes et al. (1976), with 80% of the root density in the top 40 cm of soil (Parvin et al., 2023). Lower boundary conditions were set up as a seepage face with FD or variable pressure heads describing water table levels in WT60 and WT120. The 1D domain had a 101-node discretization, divided into three layers and three observation nodes corresponding to the depth of TDR and tensiometer probes (15-, 30-, and 60-cm depth).

Bromide flux concentrations from pore-water sampling at different depths were used to calibrate solute transport parameters. The inversion was applied from March 27 to April 21, 2017 (26 days in total). The Upper boundary condition for solute transport was set as inflow concentration flux (set to 667 mg L^{-1} of Br^-), while the lower boundary condition was set as zero concentration gradient to consider purely advective transport at the outflow. To adequately simulate the real absorption of Br^- by the plants, the potential solute uptake rate value [$\text{M L}^{-2} \text{ T}^{-1}$] was adjusted to match the cumulative root solute uptake, provided by HYDRUS-1D, with the Br^- contents measured from the biomass. Both ADE and MIM equation were used to estimate solute transport parameters, such as dispersivity (λ_{ADE} and λ_{MIM}), mass-transfer coefficient (α_{MIM}), and immobile soil water content (θ_{im}). The molecular diffusion coefficient of Br^- in free water was set at $1.6 \text{ cm}^2 \text{ day}^{-1}$ (Leistra & Boesten, 2010; Lide, 2004), and Br^- was considered a conservative tracer (i.e., no sorption) due to its negligible interaction with the soil solid phase and high affinity for water (Ladu & Zhang, 2011). The maximum number of iterations was set at 10 with a number of observed Br^- concentrations in the objective function that ranged between 17 and 45.

2.8 | Goodness-of-fit indicators of model performance

To test the goodness of fit between model outcomes (S) and field observed data (O), the index of model efficiency (EF), Theil's inequality (U^2), and the determination coefficient (R^2) were calculated (Mayer & Butler, 1993; Nash & Sutcliffe, 1970; Power, 1993; Tonitto et al., 2010).

$$EF = \frac{\sum_{t=1}^N [O_{(t)} - \bar{O}]^2 - \sum_{t=1}^N [S_{(t)} - O_{(t)}]^2}{\sum_{t=1}^N [O_{(t)} - \bar{O}]^2} \quad (15)$$

$$U^2 = \frac{\sum_{t=1}^N [S_{(t)} - O_{(t)}]^2}{\sum_{t=1}^N [O_{(t)}]^2} \quad (16)$$

$$R^2 = \left\{ \frac{\sum_{t=1}^N [O_{(t)} - \bar{O}] [S_{(t)} - \bar{S}]}{\sqrt{\sum_{t=1}^N [O_{(t)} - \bar{O}]^2} \sqrt{\sum_{t=1}^N [S_{(t)} - \bar{S}]^2}} \right\}^2 \quad (17)$$

The model provides the best fit when EF , U^2 , and R^2 tend toward 1, 0, and 1, respectively. The Akaike's information criterion (AIC) was also introduced for a comparative assessment between the ADE and MIM models (Schabenberger & Pierce, 2002):

$$AIC = 2k - 2\ln(L) \quad (18)$$

where k represents the number of estimated parameters in the model, and L denotes the maximized value of the likelihood function for the model. The most effective model is the one with the lowest AIC score.

2.9 | X-ray computed microtomography

To quantify the effect of soil root apparatus on the macropore structure, a laboratory experiment was performed consisting of four reconstructed soil cores (diameter = 13.3 cm; $h = 14$ cm)—hereafter referred as Soil Core 1, 2, 3, and 4—where FR was grown for 60 days and noninvasively analyzed by X-ray microtomography (μCT). The experiment was conceived as a complementary study due to the challenges of directly investigating the complex interactions between root architecture and soil structure within the lysimeters without altering the tracer test and modeling. Its aim was to offer a tangible insight into the actual taproot growth of FR in the same soil type and its effect on soil pore size and morphology changes. The cores were filled with the same soil taken nearby the experimental site, previously sieved at 0–2 mm, and subjected to a series of wetting and drying cycles to ensure

uniform conditions and to reach a representative soil bulk density of the lysimeters (1.48 g cm^{-3} , on average). Moreover, uniform soil core conditions were sought to quantify the effect of the taproot growth on cracks and fissures formation and orientation around the roots without considering any previous difference in pore architecture that would have masked the comparison. To guarantee homogeneous growth of FR among replicates (Figure S1), each sample was equally fertilized with 0.09 g of urea, corresponding to 30 kg N ha^{-1} , that was mixed with soil just before sowing. Moreover, during the growing period, soil cores were maintained homogeneously wet at field capacity thanks to a drip irrigation system, set to 3.75 mm day^{-1} , and received photosynthetic light for 10 h each day. Samples were scanned twice at the University Grenoble Alpes with an EasyTom XL Industrial Micro & Nano Computed Tomography (RX Solutions): (1) before FR seeding (hereafter “T0”), and (2) immediately after mechanical devitalization with a knife at the plant’s collar that occurred after 60 days of growing (hereafter “T60”). The T0 was representative of BS condition, while the T60 of soil with FR of a short cycle development. The μCT operating conditions were set to 150 kV and $420 \text{ }\mu\text{A}$ using a 360° helicoid rotation inside a temperature-controlled cabinet. Beam hardening artifacts were minimized during the acquisition using a 0.5 mm copper filter. The acquisition sequence was composed of 5760 16-bit projections with a resolution of 1840×1456 voxels. The three-dimensional (3D) tomographies were reconstructed using XAct software version 2.0 (RX Solutions) to obtain a stack of 3010 two-dimensional (2D) slices in 16-bit depth. The final voxel resolution was $66 \text{ }\mu\text{m}$ in all three directions. To reduce image noise and focus the analysis on macropores only, the stack was resized to a resolution of $264 \text{ }\mu\text{m}$ by applying a mean filter. The reconstructed grayscale volumes were first converted into 8-bit grayscale depth and segmented using the Trainable Weka Segmentation machine learning algorithm (Arganda-Carreras et al., 2017), which was able to correctly identify soil matrix, soil pores, and root apparatus. Soil porosity and root 2D area and 3D volume detected by μCT were calculated by dividing the number of pore or root pixels—or voxels—by the total number of pixels—or voxels—of the analyzed area—or volume—and expressed as percentage values (%). The 2D and 3D mean pore (and root) diameters and 3D pore size classes (μm) were estimated by using a sphere-fitting method that determines the local thickness of every medial axis pixel—or voxel (Rémy & Thiel, 2002). Three-dimensional pores were divided into three classes: $264\text{--}792$, $792\text{--}1320$, and $1320\text{--}4488 \text{ }\mu\text{m}$. The Euler number—by calculating the number of redundant connections between pores per unit volume (mm^{-3})—was used as an estimate of local pore connectivity, which becomes increasingly negative with increasing connectivity and increasingly positive with decreasing connectivity (Vogel et al., 2010). The degree of anisotropy (DA) was calculated from the concepts

of mean intercept length (Harrigan & Mann, 1984). Digital image processing and analysis was done using CTAn software version 1.17.7.2 (Bruker microCT). Linear regression analyses were finally employed to explore the relationship between the growing depth (an independent variable) and root growth parameters—diameter and area—as dependent variables.

3 | RESULTS

3.1 | Weather conditions and irrigation input

During the experiment, from February 15 to April 25, 2017, the average daily temperature ranged from a minimum of 4.6°C to a maximum of 17.8°C (Figure 2). The lowest temperatures were observed in February ($<0^\circ\text{C}$) and after mid-April, when the minimum was close to 2°C . Maximum temperatures were observed between the end of March and the beginning of April, when they slightly exceeded 25°C . Episodic rainfall events occurred during the course of the experiment, with a peak of 11.6 mm day^{-1} recorded on April 4. Rainfall totally amounted 38.8 mm . The main water input for all lysimeters was provided by rainfall irrigation for a total of 290 mm (Figure 2), which was concentrated in the second half of the experimental period concurrently with the tracer test. The daily reference evapotranspiration (ET_0) varied between 0.5 and 3.2 mm day^{-1} leading to a total of 120 mm .

3.2 | Biomass yield and bromide concentrations in plants

Aboveground dry biomass of FR was on average 1702 g m^{-2} , ranging between a minimum of 1072 g m^{-2} with water table level at 60 cm (FR-WT60_a) and a maximum of 2522 g m^{-2} when it was set at 120 cm (FR-WT120_b) (Table 2). The taproot biomass developed mainly in the $0\text{--}40 \text{ cm}$ layer and weighted 185 g m^{-2} , on average, with an aboveground-to-taproot ratio of 10.17 . No significant differences were detected in either aboveground or taproot biomass concerning the water table levels.

The average bromide (Br^-) concentration in the aboveground biomass was 1721 mg kg^{-1} , with a standard error (hereafter, SE) of $\pm 198 \text{ mg kg}^{-1}$, while it reached an average of 1035 ($\text{SE} \pm 137$) mg kg^{-1} in taproots. Total Br^- uptake was 2965 ($\text{SE} \pm 220$) mg m^{-2} , on average, of which 93.5% accumulated in the aboveground biomass and only 6.5% in taproots (Table 2). Regarding the water table management, Br^- translocation to aboveground biomass was on average higher with WT120 (2965 mg m^{-2}) than WT60 (2691 mg m^{-2}); nevertheless, it was also affected by large variability that highlighted little relevance to the groundwater level with insignificant differences, being both maximum (96.3%)

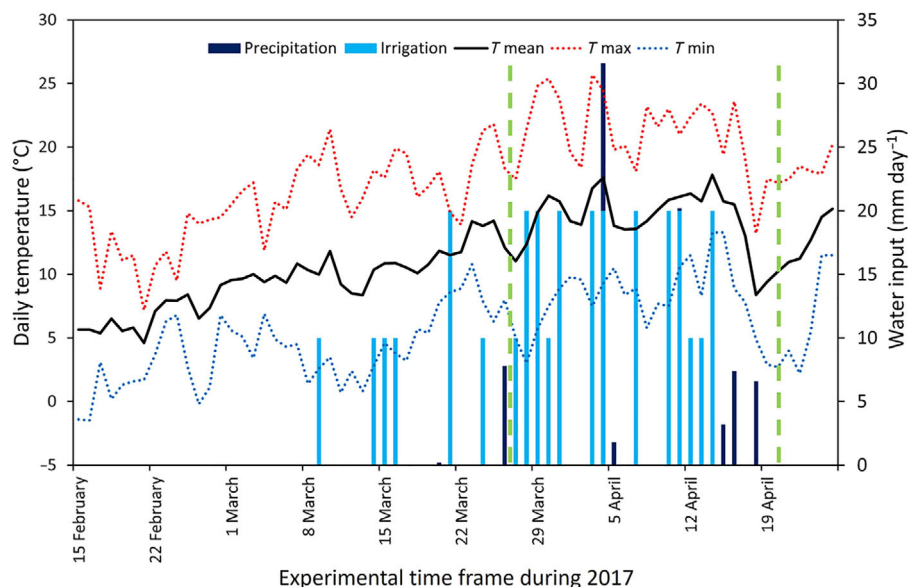


FIGURE 2 Water input (precipitation and irrigation) and average, maximum, and minimum daily temperature (T_{mean} , T_{max} , and T_{min} , respectively) recorded during the experimental period. Green dashed vertical lines indicate the start and end of the tracer test.

TABLE 2 Biomass production (dry matter), plant bromide (Br^-) concentration, and removal.

Lysimeter ID	Dry biomass			Br ⁻ concentration		Br ⁻ uptake		
	Aboveground (g m ⁻²)	Taproot (g m ⁻²)	Above-ground to taproot ratio	Aboveground (mg kg ⁻¹)	Taproot (mg kg ⁻¹)	Above-ground (mg m ⁻²)	Taproot (mg m ⁻²)	Total (mg m ⁻²)
FR-WT60a	1072	77	13.92	2593	1371	2780	106	2886
FR-WT60b	1700	290	5.86	1531	1272	2602	369	2972
FR-WT120a	1414	165	8.57	1626	1043	2299	172	2471
FR-WT120b	2522	246	10.25	1440	1001	3632	246	3878
FR-FDb	1800	147	12.24	1414	489	2545	72	2617

Note: Lowercase letters denote the two replicates within a single treatment.

Abbreviations: FD, free drainage; FR, forage radish; WT60, groundwater at 60-cm depth; WT120, groundwater at 120-cm depth.

and minimum (87.6%) accumulation in the aboveground biomass found under the shallower groundwater condition (FR-WT60). Compared to the amount initially distributed, that is, 6667 mg m⁻², FR adsorbed 39.3% in FD, 43.9 ± 0.6% in WT60, and 47.6 ± 10.6% in WT120 over the 30-day growing period.

3.3 | Soil water dynamics

3.3.1 | Experimental soil water content and pressure heads

Measured volumetric soil water content (VWC) within the soil profile showed higher values in the BS treatments (Figure 3A–E) than in the FR treatments (Figure 3F–L), with differences that were more emphasized in the topsoil (+15%)

than in the deepest layers, where +8% and +3% were recorded at 30- and 60-cm depth, respectively. In general, VWC was lower in top layers than deeper ones, with more fluctuations due to irrigation and rainfall inputs that were partly offset by evapotranspiration and drainage. Minimum topsoil values were observed at the beginning of the experiment, when average VWC was 0.12, 0.16, and 0.20 m³ m⁻³ in FD, WT120, and WT60, respectively. Instead, peaks were found on different days according to the peculiar characteristics of each lysimeter. For instance, under BS-WT60 (Figure 3A,B), close-to-saturation conditions were observed just 2 weeks after the first watering (40 days after the beginning of the experiment), which persisted in the following 20 days. In contrast, maximum values in FR-WT60 (Figure 3F,G) were found about 10 days later, after 32 mm rainfall, followed by a sharp decrease due to water drainage and evapotranspiration processes. In WT60, negligible soil moisture differences between BS and

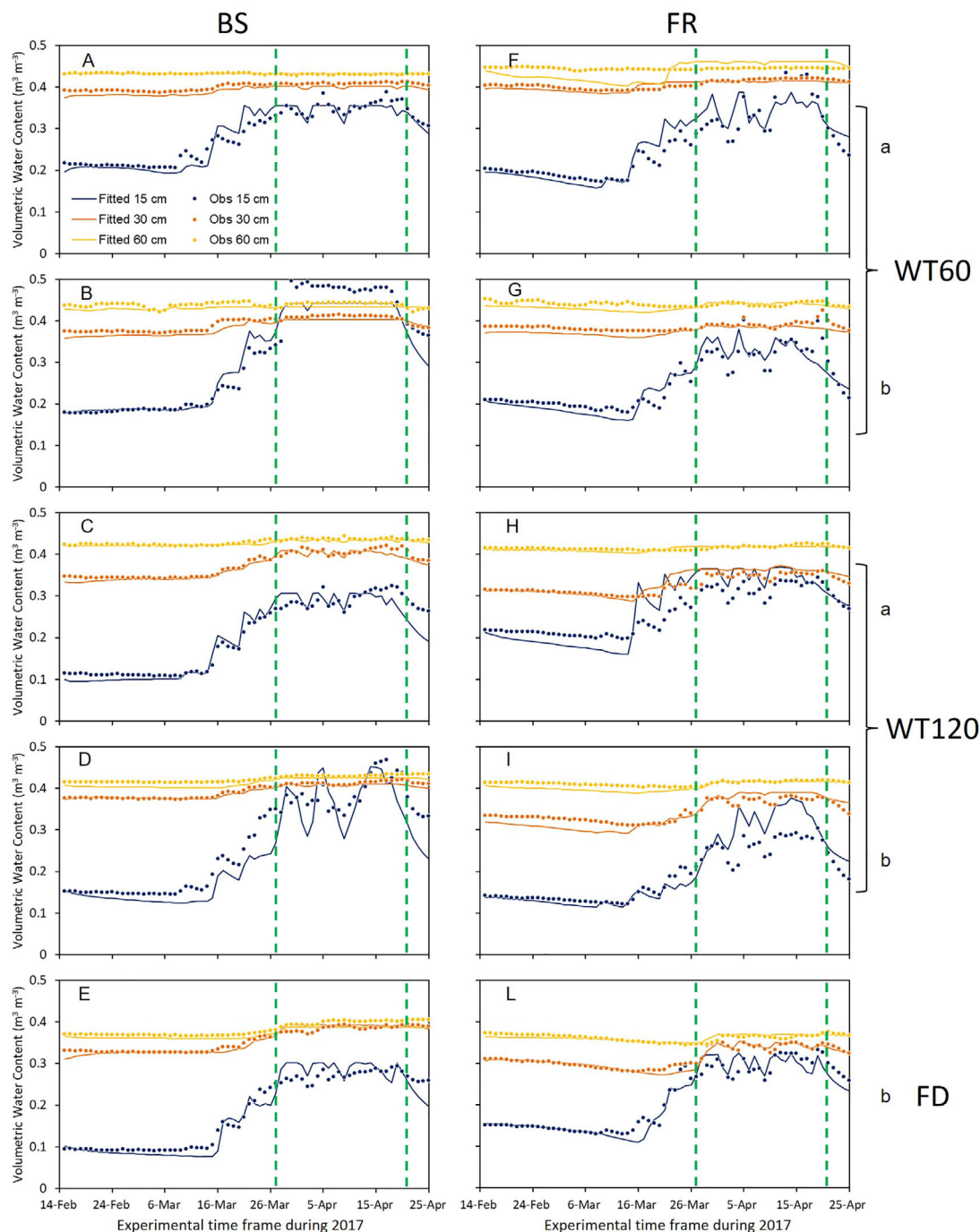


FIGURE 3 Observed (“Obs” in the legend) and predicted (“Fitted” in the legend) soil volumetric water contents at different soil depths (15, 30, and 60 cm) recorded during the experimental period. FD, free drainage; FR, forage radish; BS, bare soil; WT60, groundwater at 60-cm depth; WT120, groundwater at 120-cm depth. The “a” and “b” letters denote the two replicates within a single treatment. Green dashed vertical lines indicate the start and end of the tracer test.

FR at both 30 and 60 cm (0.41 and $0.44 \text{ m}^3 \text{ m}^{-3}$ on average, respectively) were observed. In WT120 and particularly in FD, lower VWC was recorded with FR compared with BS, even at the deepest depth.

The recorded pressure head (Figure 4) followed VWC dynamics, showing a differentiation between the first half of

the experimental period, before the first watering, and the second half, when intense watering occurred. Following the onset of high-frequency watering during Br^- testing, the pressure head rose rapidly in all lysimeters, regardless of the water table management and the presence of FR, reaching everywhere values above -50 cm .

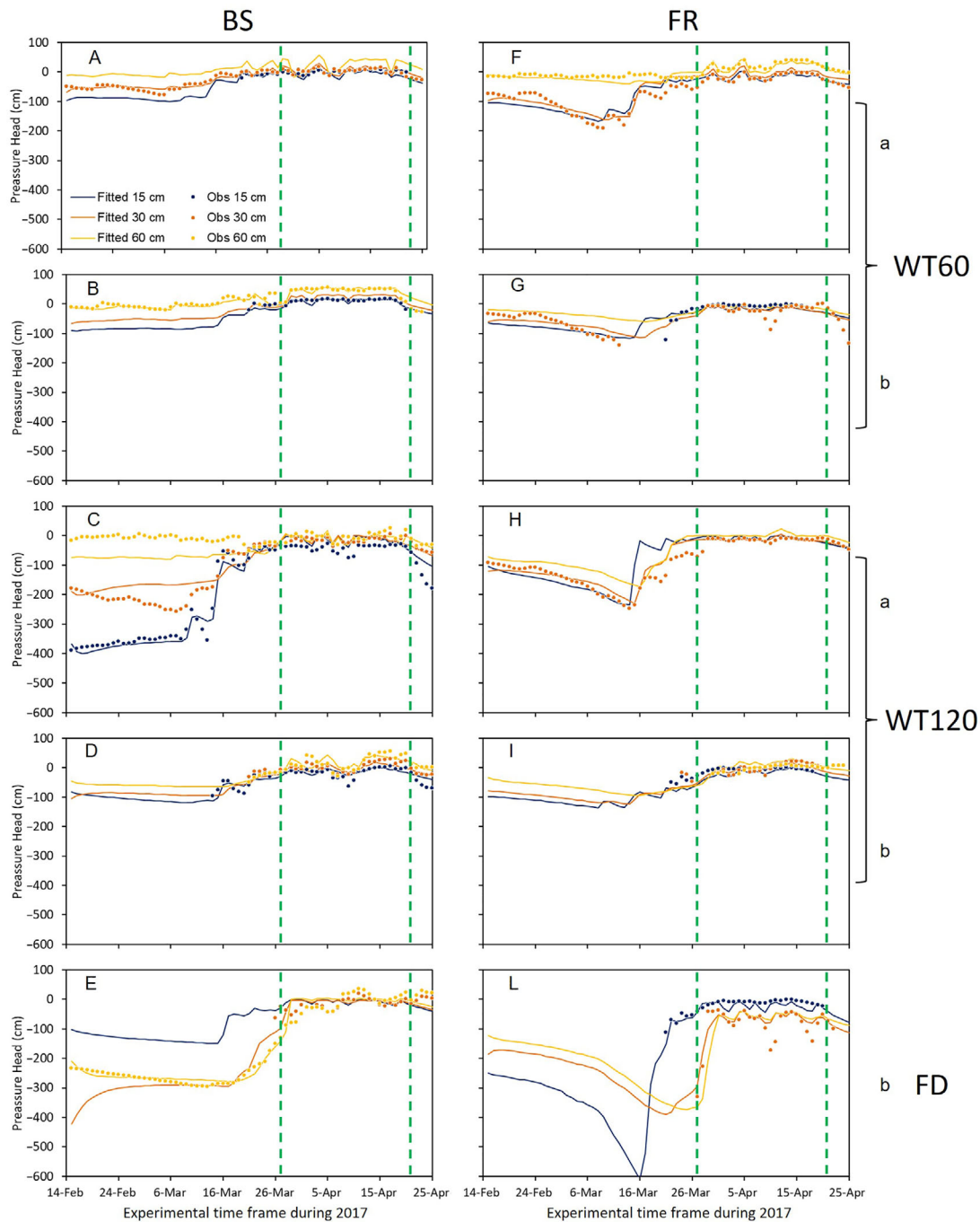


FIGURE 4 Observed (“Obs” in the legend) and predicted (“Fitted” in the legend) pressure head at different soil depths (15, 30, and 60 cm) recorded during the experimental period. BS, bare soil; FD, free drainage; FR, forage radish; WT60, groundwater at 60-cm depth; WT120, groundwater at 120-cm depth. The “a” and “b” letters denote the two replicates within a single treatment. Green dashed vertical lines indicate the start and end of the tracer test.

3.3.2 | Modeled soil water content, pressure heads, and parameter estimation

Model inversion estimated well the soil water dynamics of all lysimeters, quantified by very high model efficiency ($EF = 0.946$, on average) and low inequality ($U^2 = 0.003$, on average) between observed and modeled values (Table 1S).

The goodness of fit of simulated versus observed pressure heads was satisfactory (Table 1S), except for lysimeter FR-WT120_b. The average EF and U^2 were 0.707 and 0.223, respectively. Likewise, mirroring the dynamics of soil water content, the model accurately predicted the increase in pressure head after the initial watering in all lysimeters; additionally, it successfully captured the near-saturation values

TABLE 3 Soil hydraulic parameters estimated with HYDRUS-1D at different layer depths, under forage radish (FR) and bare soil (BS) treatments with different groundwater management conditions (groundwater at 60-cm depth [WT60], groundwater at 120-cm depth [WT120], and free drainage [FD]).

Lysimeter ID	Layer (cm)	θ_s (m m ⁻³)		α (cm ⁻¹)		n		K_s (cm day ⁻¹)	
		Value	SE	Value	SE	Value	SE	Value	SE
BS-WT60a	0–20	0.356	±0.004	0.023	±0.001	1.66	±0.05	5.1	±0.2
	20–45	0.402	±0.003	0.013	±0.000	1.13	±0.01	4.4	±0.1
	45–130	0.432	±0.004	0.001	±0.000	1.10	±0.05	33.9	±2.5
BS-WT60b	0–20	0.442	±0.010	0.049	±0.002	1.61	±0.06	16.4	±0.3
	20–45	0.403	±0.006	0.022	±0.000	1.15	±0.04	10.1	±0.2
	45–130	0.433	±0.006	0.011	±0.001	1.15	±0.05	2.3	±0.0
BS-WT120a	0–20	0.308	±0.005	0.017	±0.001	1.65	±0.03	17.4	±1.3
	20–45	0.409	±0.004	0.012	±0.000	1.19	±0.01	2.3	±0.2
	45–130	0.436	±0.004	0.005	±0.000	1.15	±0.03	9.2	±0.8
BS-WT120b	0–20	0.452	±0.011	0.065	±0.003	1.66	±0.05	44.9	±3.5
	20–45	0.410	±0.007	0.010	±0.001	1.16	±0.04	21.4	±4.9
	45–130	0.425	±0.007	0.010	±0.000	1.14	±0.02	1.8	±0.1
BS-FDb	0–20	0.302	±0.004	0.032	±0.001	1.94	±0.00	1.0	±0.0
	20–45	0.393	±0.002	0.006	±0.000	1.21	±0.01	3.6	±0.1
	45–130	0.388	±0.001	0.003	±0.000	1.15	±0.01	1.4	±0.0
FR-WT60a	0–20	0.388	±0.006	0.033	±0.001	1.54	±0.02	16.8	±0.7
	20–45	0.415	±0.004	0.005	±0.000	1.16	±0.03	92.2	±6.8
	45–130	0.461	±0.005	0.079	±0.003	1.11	±0.01	7.9	±0.4
FR-WT60b	0–20	0.398	±0.009	0.053	±0.006	1.51	±0.04	5.3	±1.0
	20–45	0.392	±0.005	0.014	±0.003	1.10	±0.02	53.4	±9.1
	45–130	0.445	±0.005	0.015	±0.004	1.10	±0.03	10.9	±1.7
FR-WT120a	0–20	0.371	±0.003	0.037	±0.001	1.40	±0.02	1.5	±0.1
	20–45	0.373	±0.006	0.018	±0.001	1.17	±0.01	68.5	±0.5
	45–130	0.419	±0.002	0.003	±0.000	1.10	±0.01	1.7	±0.1
FR-WT120b	0–20	0.377	±0.014	0.044	±0.008	1.70	±0.09	204.2	±31.3
	20–45	0.390	±0.008	0.016	±0.003	1.32	±0.08	16.8	±2.2
	45–130	0.416	±0.010	0.006	±0.000	1.20	±0.03	0.5	±0.0
FR-FDb	0–20	0.331	±0.003	0.023	±0.000	1.44	±0.02	1.2	±0.0
	20–45	0.370	±0.006	0.010	±0.000	1.21	±0.02	18.3	±0.6
	45–130	0.373	±0.002	0.003	±0.000	1.10	±0.01	30.9	±4.2

Note: Lowercase letters in Lysimeter ID column denote the two replicates within a single treatment. Standard errors (±SE) are also reported.

that occurred during the latter half of the experimental period.

Calibrated water retention parameters showed differences between the treatments at the three depths (Table 3). The saturated soil water content (θ_s) averaged 0.39 and 0.40 m³ m⁻³ in the FR and BS, with minimum and maximum values of 0.32 and 0.46 m³ m⁻³, and 0.30 and 0.45 m³ m⁻³, respectively. Larger differences of θ_s were observed between groundwater managements with higher values in WT60 (0.41 m³ m⁻³, on average) and WT120 (0.40 m³ m⁻³, on average) than FD (0.36 m³ m⁻³, on average).

The average α parameter was 0.024 cm⁻¹ in FR and 0.019 cm⁻¹ in BS, ranging between minimum and maximum

values of 0.001–0.079 and 0.003–0.065 cm⁻¹, respectively. In contrast, a tendency to soil layering was observed, with higher values at 0–20 cm (0.037 cm⁻¹) and lower values at 20–45 cm and 45–130 cm, 0.013 and 0.014 cm⁻¹, respectively. Similarly, the average n parameter decreased from the top (1.61) to the deeper layers (1.18 and 1.13 at 20–45 cm and 45–130 cm, respectively). This decreasing trend of α and n with depth is consistent for all the tests reflecting an increase of air entry pressure and a decrease of the retention curve stiffness. This result appeared consistent with the increase of the finer granulometric fractions (silt and clay) with depth (Table 1).

Finally, an increase of saturated soil hydraulic conductivity (K_s) was observed in FR compared to BS in the 20–45 cm

layer (49.8 cm day^{-1} vs. 8.4 cm day^{-1} on average, respectively), while the high variability at the surface layer did not highlight clear differences between treatments.

3.4 | Solute dynamics

3.4.1 | Experimental bromide movement and detection

Bromide concentration in the soil pore water (Br^- , mg L^{-1}) was never found before the starting of the experiment at the different depths (i.e., below the detection limit of 0.1 mg L^{-1}). After application, Br^- showed different dynamics as related to sampling depth and soil cover. In BS, high Br^- concentrations were detected at 15-cm depth just 1 day after distribution, reaching peaks of 61.3, 75.9, and 79.8 mg L^{-1} in WT60, WT120, and FD, respectively. Thereafter, Br^- decreased almost linearly, until values $<10 \text{ mg L}^{-1}$, 18 days after Br^- distribution (Figure 5A–E).

At 30-cm depth, detection of maximum solute concentration was 8 days after Br^- distribution under shallow water tables (WT60 and WT120) ranging between 8.8 and 51.9 mg L^{-1} , while the peak was reached only after 2 days in FD following similar dynamics to topsoil observations. Finally, a slow and steady increase of Br^- was observed at 60 cm, reaching a maximum 16 days after tracer distribution, varying from 8.3 mg L^{-1} in WT60 to 26.6 mg L^{-1} in FD.

Compared to BS, distinct differences in tracer dynamics were observed in FR, where the impact of vegetation's uptake reduced Br^- peak concentrations in the upper layer by at least half (Figure 5H,I,L). In fact, Br^- concentration exceeded 60 mg L^{-1} only at the topsoil of one of the FR-WT60 replicate (Figure 5G) 2 days after application. Afterwards, Br^- sharply decreased until values that never exceeded 5 mg L^{-1} , 4 days later. The effect of FR was also found at increasing depths, since Br^- never exceeded 6.8 and 2.0 mg L^{-1} at 30 and 60 cm, respectively, throughout the experiment.

3.4.2 | Modeled solute transport dynamics and parameterization

Solute transport under different treatments was effectively characterized by both ADE and MIM models (Figures 5 and 6). For instance, the pronounced Br^- peak at a depth of 15 cm, followed by a gradual decrease until the end of the simulation, in the BS, as well as the dissimilar forms noticed in WT60 compared to WT120 and FD under FR, were reasonably well simulated.

The fitting showed generally some slight improvement with MIM equation than ADE, regardless of the groundwater and soil management (Table 2S). Only in one case, BS-WT60_b,

both models yielded unsatisfactory results, with negative model efficiency EF (-1.42 with ADE and -0.32 with MIM equation) and high inequality U^2 (1.38 with ADE; 0.76 with MIM equation). Note that AIC was generally lower with MIM than ADE simulations, emphasizing an effective improvement of solute dynamic description with nonequilibrium models.

In a few treatments, exclusively in the FR lysimeters, the scarce availability of experimental data (Figure 6H,I,L) introduced some uncertainty in model parameterization of the deepest layer (45–140 cm). In such cases, HYDRUS-1D simulated a quick Br^- uptake that reached the peak 2 days after distribution (Figure 7), and thereafter a decrease driven by the lower Br^- concentration available in the pore water. It resulted in a strong reduction of Br^- leaching below 40-cm depth in FR compared to BS treatments, as exemplified in Figure 8. It follows that only solute transport results obtained from surface and intermediate layers were further considered (i.e., 0–20 and 20–45 cm) (Table 4).

Very high dispersivity (λ_{ADE}) was found in certain lysimeters at various depths when applying the ADE model. In some cases, the calculated λ_{ADE} values exceeded the corresponding layer thickness. Large values of λ_{ADE} indicated most likely nonequilibrium conditions (Radcliffe & Šimůnek, 2018), making preferable the use of MIM model. The MIM-estimated dispersivity (λ_{MIM}) was consistently smaller than the layer thickness, which appeared more physically consistent than values obtained with ADE (Table 4). It showed negligible differences between BS and FR in the topsoil layer with average values of 3.34 and 2.72 cm, respectively. In the 20–45 cm layer, the dispersivity increased on average for BS, while it remained comparable to the layer above for FR with values of 5.13 and 2.44 cm, respectively. This outcome should be interpreted considering the significant disparities observed in the immobile water content (θ_{im}) between FR and BS. Specifically, the values in FR were, on average, 21% lower than those in BS. Finally, the degree of solute exchange between the immobile and mobile regions, α_{MIM} (day^{-1}), was similar between BS and FR, especially in the surface layer.

3.5 | X-ray computed microtomography

After 60 days of FR growth (T60), the total macro-porosity ($\geq 264 \mu\text{m}$) increased by 250% from $0.28 \pm 0.07\%$ to $0.70 \pm 0.09\%$ (Table 5). Similarly, the mean pore diameter increased from $623 \mu\text{m} \pm 29$ to $1110 \mu\text{m} \pm 108$, regardless of some difference in the soil bulk density between samples. Soil macro-porosity at seeding (T0) was dominated by pores $<792 \mu\text{m}$, whereas 792–1320 μm and 1320–4488 μm pore classes contributed only $<15\%$ and 1.3% , respectively. Conversely, at T60, the 792–1320 μm pore class represented 31% of the total macro-porosity, and similarly, the 1320–4488 μm pore class increased to an average of 28%. The pore number

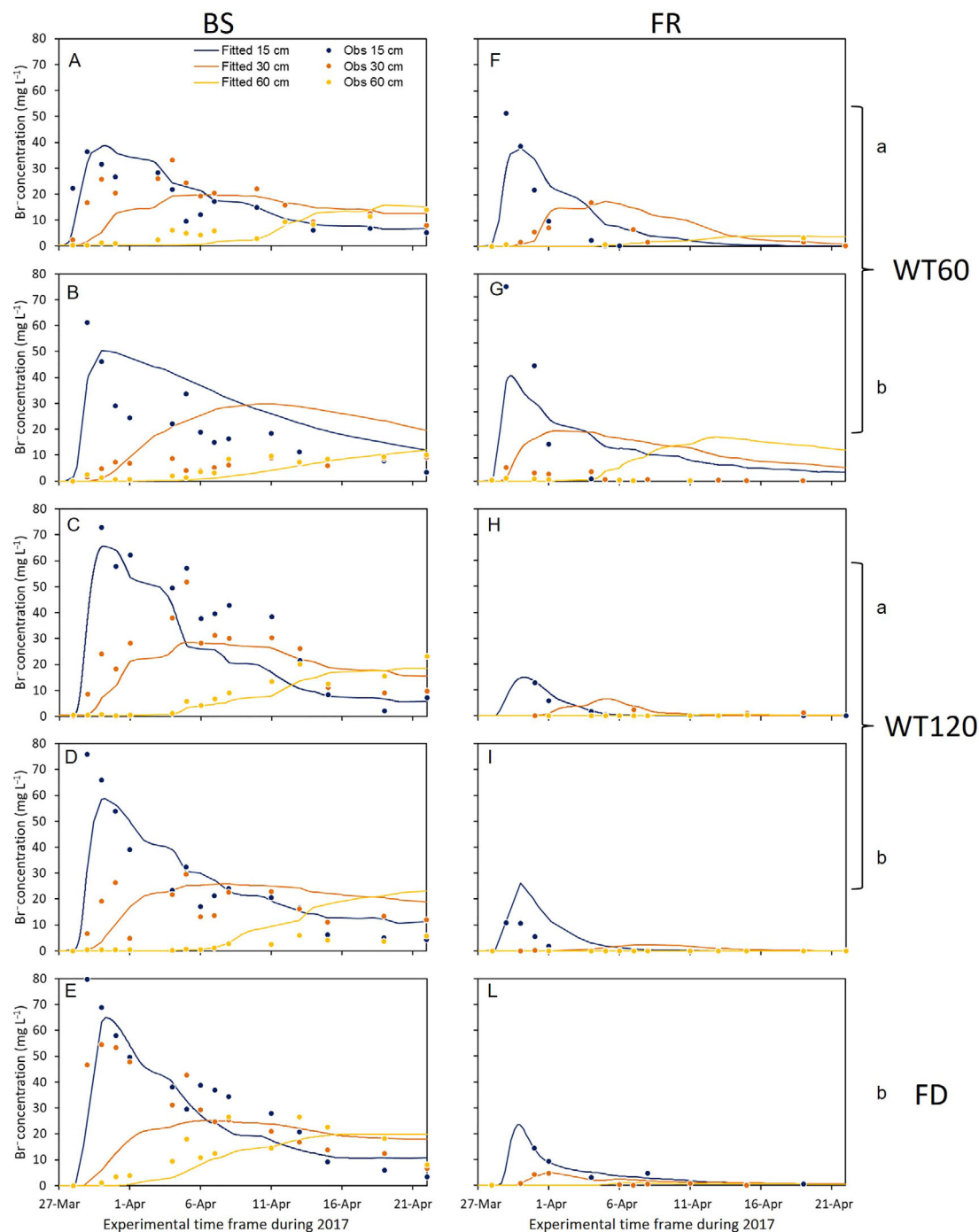


FIGURE 5 Experimental Br⁻ concentration (“Obs” in the legend) and simulated (“Fitted” in the legend) dynamics obtained by solute modeling applying advection–dispersion equation (ADE) at different soil depths (15, 30, and 60 cm) recorded during the experimental period. BS, bare soil; FD, free drainage; FR, forage radish; WT60, groundwater at 60-cm depth; WT120, groundwater at 120-cm depth. The “a” and “b” letters denote the two replicates within a single treatment.

and the Euler number showed less variability between scan times, while the degree of anisotropy decreased after the FR growth (Table 5).

At T60, average values of soil porosity at the surface exceeded 4%, while at T0 they reached a maximum of only 1.5%. No relevant changes were observed deeper than 5 cm.

Similarly, the 2D mean pore diameter increased at T60 (average of 851 μm) compared to T0 (average of 268 μm) up to a depth of 5 cm, while it was less affected by root growth at greater depth (average = 414 μm vs. 345 μm , respectively) (Figure 9B). Root diameter along the soil profile significantly decreased from about 10 mm at the surface to <2 mm at

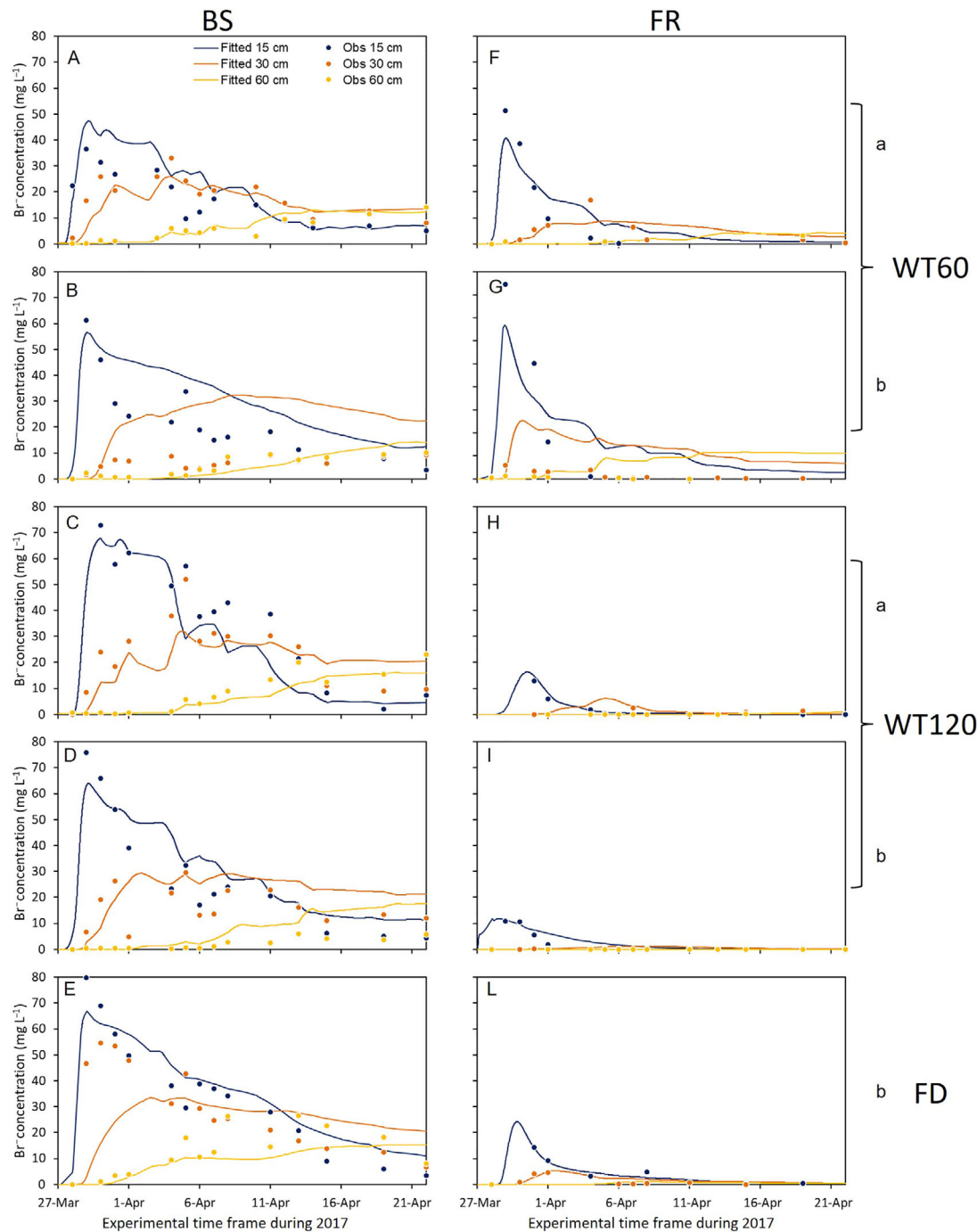


FIGURE 6 Experimental Br^- concentration (“Obs” in the legend) and simulated (“Fitted” in the legend) dynamics obtained by solute modeling applying mobile-immobile (MIM) equation at different soil depths (15, 30, and 60 cm) recorded during the experimental period. FD, free drainage; FR, forage radish; BS, bare soil; WT60, groundwater at 60-cm depth; WT120, groundwater at 120-cm depth. The “a” and “b” letters denote the two replicates within a single treatment

6-cm depth ($R^2 = 0.57$; $p < 0.001$), while no roots were detected with μCT below 10-cm depth (Figure 9D). Similarly, the proportion of root area to total area decreased (Figure 9C). Finally, a qualitative view of 2D sections and a 3D reconstruction after segmentation emphasized the presence of radial fissures and pores around the taproot apparatus at T60 (Figure 10; see also Figures S2–S4).

4 | DISCUSSION

The FR growth in the lysimeters has substantially modified the water and solute dynamics compared to BS conditions. FR acted as an excellent catch crop by adsorbing at least 39% of the total amount of injected tracer and transferring most of it in aboveground biomass, thereby reducing Br^- pore water

TABLE 4 Dispersivities (λ_{ADE} and λ_{MIM}), immobile water content (θ_{im}), and mass exchange coefficient (α_{MIM}) estimated using advection–dispersion equation (ADE) and mobile–immobile (MIM) equation at different layer depths, under forage radish (FR) and bare soil (BS) treatments with different groundwater management conditions (groundwater at 60-cm depth [WT60], groundwater at 120-cm depth [WT120], and free drainage [FD]).

Lysimeter ID	Layer (cm)	λ_{ADE} (cm)		λ_{MIM} (cm)		θ_{im} (m ³ m ⁻³)		α_{MIM} (day ⁻¹)	
		Value	SE	Value	SE	Value	SE	Value	SE
BS-WT60a	0–20	7.42	±1.08	1.76	±0.43	0.312	±0.015	0.23	±0.03
	20–45	13.98	±5.73	5.23	±2.26	0.395	±0.011	0.04	±0.02
	45–130	0.13		1.76		0.430		0.29	
BS-WT60b	0–20	10.60	±2.30	6.12	±0.94	0.289	±0.046	0.17	±0.06
	20–45	5.20	±3.63	0.14	±0.05	0.395	±0.007	0.09	±0.06
	45–130	30.00		1.74		0.431		0.09	
BS-WT120a	0–20	4.34	±0.83	1.46	±0.48	0.240	±0.023	0.28	±0.07
	20–45	14.55	±4.48	7.48	±2.75	0.390	±0.013	0.18	±0.12
	45–130	4.28		3.60		0.313		0.84	
BS-WT120b	0–20	5.03	±0.75	3.68	±0.80	0.250	±0.012	0.18	±0.04
	20–45	76.12	±32.70	4.99	±2.47	0.402	±0.002	0.09	±0.05
	45–130	0.00		0.00		0.421		0.21	
BS-FDb	0–20	4.67	±0.34	3.65	±0.65	0.256	±0.014	0.17	±0.03
	20–45	2359.30	±295.98	7.81	±4.09	0.366	±0.120	0.04	±0.03
	45–130	3.90		8.53		0.373		0.05	
FR-WT60a	0–20	6.43	±0.90	3.37	±0.69	0.223	±0.043	0.15	±0.03
	20–45	1.72	±1.38	3.78	±1.60	0.280	±0.103	22.40	±5.60
	45–130	35.96		0.72		0.450		0.14	
FR-WT60b	0–20	10.11	±2.87	5.13	±1.49	0.269	±0.016	0.12	±0.04
	20–45	55.89	±24.94	6.53	±5.62	0.378	±0.037	0.10	±0.15
	45–130	0.01		0.16		0.432		0.23	
FR-WT120a	0–20	4.68	±0.37	1.49	±0.33	0.121	±0.021	0.02	±0.01
	20–45	0.01	±0.00	0.01	±0.00	0.054	±0.019	0.01	±0.04
	45–130	49.46		0.13		0.113		7.36	
FR-WT120b	0–20	2.19	±0.39	2.12	±1.12	0.180	±0.021	0.28	±0.04
	20–45	0.02	±3.17	0.04	±0.04	0.190	±0.176	0.55	±1.21
	45–130	0.02		0.70		0.075		0.09	
FR-FDb	0–20	25.55	±3.64	1.47	±0.30	0.247	±0.020	0.07	±0.01
	20–45	3.58	±1.23	1.86	±0.72	0.271	±0.077	0.03	±0.03
	45–130	0.06		0.04		0.349		0.01	

Note: The letters “a” and “b” denote the two replicates within a single treatment. Parameterization of the deepest layer was obtained by model optimization carried out on homogeneous soil profile, only, due to poor availability of experimental data. Standard errors (±SE) are also reported.

concentration from the surface layer to the greatest depth (60 cm), where Br[−] never exceeded 1.3 mg L^{−1} regardless of groundwater depth. The ability to adsorb water and solutes, even more than other crops, was a confirmation of previous studies such as those of Wahlström et al. (2015) and Baggs et al. (2000), the latter reporting in a sandy loam/loamy sand soil a reduction in nutrients leaching of 44% compared with a fallow field.

A change in the pore architecture with implications on fluxes through the soil can also occur, as reported by Abdollahi et al. (2014), that found an increase in continuous

macropores and enhanced air permeability down to 27-cm depth in a comparison between sandy loam soils managed with FR and without. Despite the fact that our core experiment was conducted with soils of different texture (silt loam) compared to Abdollahi et al. (2014), the μ CT analysis still provided evidence of augmented total macroporosity (resolution = 264 μ m) and mean pore diameter (>792 μ m) after 60 days of plant growth, which was clearly associated with the taproot effect on soil structure rearrangement as likely happened in the lysimeters. However, in the cores, the effect was only in the top 6 cm (Figures 9 and 10) because a relatively

TABLE 5 Soil bulk density and X-ray microtomography (μ CT)-derived parameters of the four reconstructed soil cores at scan time T0 (bare soil) and T60 (60 days forage radish [FR] growth).

Soil properties	Soil Core 1		Soil Core 2		Soil Core 3		Soil Core 4	
	T0	T60	T0	T60	T0	T60	T0	T60
Bulk density (g cm^{-3})	1.43	1.43	1.44	1.44	1.54	1.54	1.53	1.53
Total macro-porosity (%)	0.34	0.75	0.48	0.96	0.20	0.66	0.10	0.44
Mean pore diameter (μm)	541	940	684	1386	672	1226	593	846
Pore size class 264–792 μm (%)	98	53	74	28	73	28	89	56
Pore size class 792–1320 μm (%)	2	28	22	30	27	34	10	32
Pore size class 1320–4488 μm	0	19	4	42	0	38	1	12
Number of pores (–)	4461	3604	2684	2494	1103	1301	1397	4047
Euler number ($\mu\text{m}^{-3} \times 10^{-5}$)	8.09	8.17	7.07	5.53	1.80	2.47	1.90	4.37
Degree of anisotropy (–)	0.71	0.42	0.80	0.42	0.81	0.45	0.69	0.36

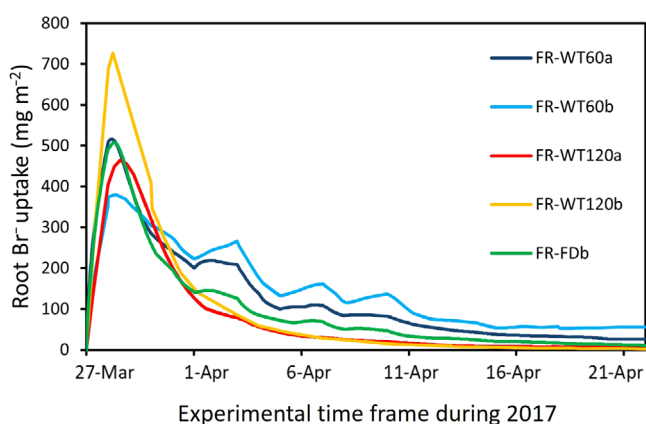


FIGURE 7 Modeled root Br^- uptake (mg m^{-2}) by forage radish during the tracer test in the FR lysimeters. FD, free drainage; FR, forage radish; WT60, groundwater at 60-cm depth; WT120, groundwater at 120-cm depth. The “a” and “b” letters in the legend denote the two replicates within a single treatment.

shallow root penetration was due to indoor cultivation in the small containers compared with growth dynamics in open-field conditions. Bodner et al. (2014) found similar results in sandy clay loam soil, highlighting that a change in soil structure was strongly dependent on the root system, regardless of the soil type. In fact, they reported that fine rooting systems, which tended to use already existing growth paths, increased the microporosity. In contrast, plants with coarse roots created new growth paths, which were associated with an increase of macroporosity. The same authors suggested that in the presence of soil penetration by a taproot, as in the case of our FR, a reorientation of soil particles may occur due to a root-enhanced mechanical resistance that counteracts a tight packing between aggregate surfaces, thereby increasing the inter-aggregate void space mediated by roots (Hallett et al., 2022).

The effects on pore structure arrangement, as observed and quantified with μ CT analysis in the soil cores, were also

partially reflected on soil hydraulic parameters estimated by inverse modeling with HYDRUS-1D at the lysimeter scale. The model HYDRUS-1D was effectively able to predict the sharp increase in VWC that was found at -15 cm under FR versus BS and under different groundwater managements, just after the water input events starting from March 16. Furthermore, it described correctly the VWC dynamics all along the soil profiles, both when the water table (both WT60 and WT120) has led to prolonged saturation conditions, and the FD with FR maintained $\text{VWC} < 0.38 \text{ m}^3 \text{ m}^{-3}$ along the soil profile. Note that, especially in BS, VWC was often close to saturation at 30 cm due to conditions of surface evaporation that was compensated by the high upward flux of a shallow water table, which was combined with the peculiarity of our silty loam soils of relatively high air-entry suction, even below 60 cm (Mencaroni et al., 2021; Morari, 2006). In this context, similar values irrespective of the treatments were found for both α (the inverse of air-entry suction) and the saturated water content (θ_s). Conversely, the n and saturated hydraulic conductivity (K_s) parameters changed according to the different treatments. The n parameter, which describes the pore size distribution, was lower in FR compared to BS in the top layer, suggesting a broader distribution of the pore size classes (Gabriel et al., 2019) as a consequence of the macropore development. The FR also affected K_s , which was higher in the 20–45 cm layer than under BS conditions. These results were in accordance with previous studies (García-González et al., 2018; Haruna et al., 2018) that related higher K_s to the reduction of occluded porosity during cover crop growth (Villamil et al., 2006), suggesting the likely improvement of root-mediated pore connectivity as well as the creation of channeling flows along living roots. Our noninvasive X-ray μ CT imagery, despite being conducted in relatively small PVC cylinders hosting just one taproot growth as compared to the lysimeter scale, appears to contradict previous findings. Differences in the Euler number, which could confirm enhanced pore connectivity due to root growth, were not observed, most

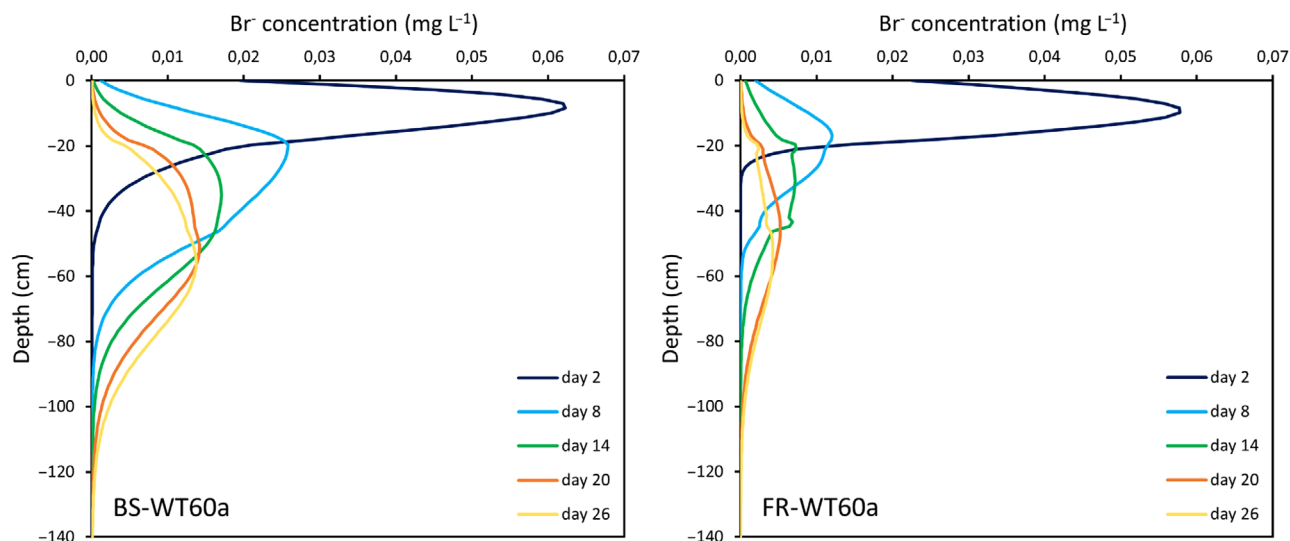


FIGURE 8 Modeled trend of Br^- concentration (mg L^{-1}) along the soil profile at five subsequent time days during the tracer test (days 2, 8, 14, 20, and 26 after distribution). Comparison between a bare soil lysimeter (on the left, BS-WT60a) and a forage radish lysimeter (on the right, FR-WT60a). BS, bare soil; FR, forage radish; WT60, groundwater at 60-cm depth.

likely because the μCT scanning suffered from low resolution. On the contrary, this analysis allowed us to verify the presence of large fissures existing along and around the roots (Figure 10), which likely developed also in the lysimeters, enhancing infiltration and promoting nonequilibrium flow. Note that the degree of anisotropy did not allow for the observation of an increase in pore vertical alignment due to the taproot; in fact, it decreased in the FR treatment compared to the BS treatment due to both vertical and radial fissures around the taproot. An analogous increase in K_s was observed in the same soil as ours by Sartori et al. (2022), where they compared soil covered with FR to BS. The study involved measurements right after devitalization of the FR, mirroring conditions to the current lysimeter experiment. The approach ensured resemblance before root decay could create distinct holes (White & Weil, 2011).

Concerning groundwater management, negligible differences were observed in soil water retention properties and K_s between FD and shallow water tables at different depths. The sole distinction identified was the higher θ_s in WT60 and WT120, as opposed to FD, within the soil profile. Mencaroni et al. (2021) confirmed these findings in study involving Br^- infiltration on the same lysimeters. Their nondestructive analyses of soil structure, performed with μCT at fine resolution ($27.3 \mu\text{m}$), highlighted a slight increase in macropore volume in the treatments exposed to shallow water table. The authors also stressed the need to use nonequilibrium models (e.g., MIM) for the most accurate representation of solute transport, even when dealing with a shallow water table (Mencaroni et al., 2021). Indeed, the authors discovered that when modeling Br^- movement using physical equilibrium models (i.e., ADE), the resulting dispersivity (λ_{ADE}) frequently exceeded

the transport distance suggesting the occurrence of preferential solute movement (Mencaroni et al., 2021; Vervoort et al., 1999).

In our experiment, we also observed some lysimeters in which λ_{ADE} similarly exceeded the thickness of the corresponding layer, but without any effect due to the water table level. Moreover, a consistent improvement in simulating solute movement using MIM equation compared to ADE was noted, regardless of λ_{ADE} values. Notably, based on the AIC comparison, this improvement could not be attributed to the higher number of variables in MIM equation compared to ADE.

Our inversion showed no differences between FR and BS with respect to dispersivity (λ_{MIM}) and mass exchange coefficient (α_{MIM}), which exhibited similar values between the two treatments. Conversely, differences were observed in immobile water content (θ_{im}) in the 0–45 cm layer, with consistently lower values in FR than in BS and, in turn, larger mobile water in FR. This result could probably be attributed to the effect of the root system on the soil pore structure providing a better access of solute to the finer part of the porosity. The μCT seems to corroborate such results, in which macropores enhanced even during the active FR taproot growth, and not only after its death and the taproot decomposition. If MIM is indeed a physically relevant model for simulating solute transport, it is plausible that the fraction of mobile water could be a proxy of the inter-aggregate pores and macropores fraction, as proposed by Batany et al. (2019). Consequently, the presence of a taproot appears to correlate with the greater presence of macroporosity (Blanco-Canqui & Ruis, 2020). Nevertheless, this increase in macroporosity in FR, along with a modest increase in K_s , did not result in noteworthy

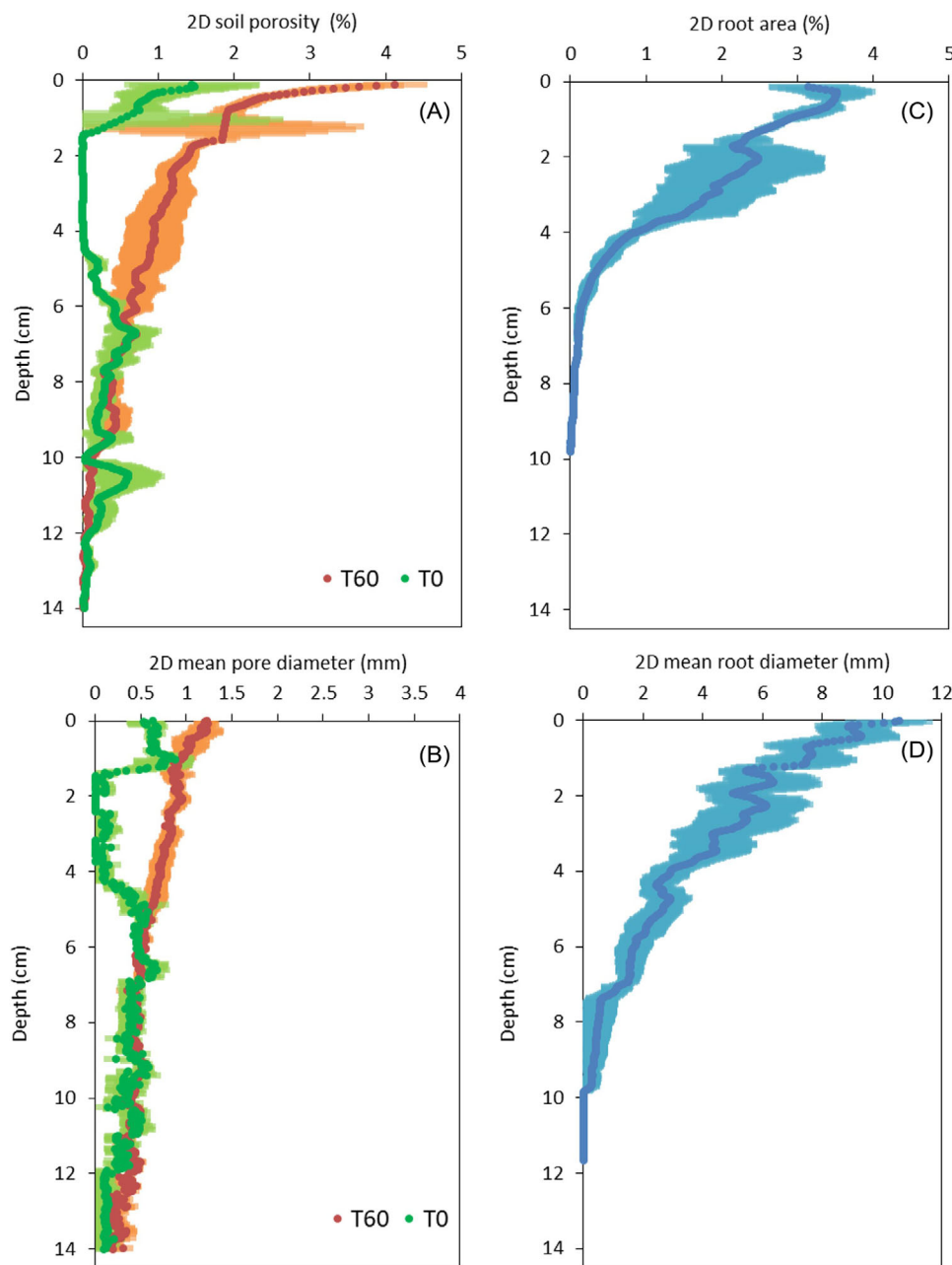


FIGURE 9 Two-dimensional (2D) distribution along the core profile of soil porosity (A, percentage value of total 2D area) and mean pore diameter (B, μm), and root area (C, percentage value of total 2D area) and mean root diameter (D, μm) by means of X-ray microtomography (resolution = 264 μm). Average values (dark dotted) along with standard errors (areas with light shades) are reported ($n = 4$).

leaching beyond the taproot system. This underscores the pivotal function of living roots in alleviating water contamination by diminishing solute percolation and assimilating solutes from pore water. This role is clearly emphasized by the simulation of Br^- uptake by plants using HYDRUS-1D.

In contrast, FR can hardly survive during winter in temperate regions and then quickly degrades and generates wider open bio-channels at the soil surface, suggesting that the deep release of solutes may occur. With these scenarios, FR could act as a driving, instead of limiting, factor to increase the leaching of nutrients and contaminants to groundwater. It

is noteworthy that this study was conducted at the lysimeter scale, which did not represent the 3D water and solute dynamics, suggesting the need of upscaling under real field conditions to validate the insights of our study. In contrast, lysimeters are designed as mesocosms capable of controlling external factors and therefore can help address some specific questions (e.g., effect of water table level, vegetation growth under field-like conditions) even if with a limited number of replicates. This constraint, nevertheless, could be crucial to consider in designing future experiments aimed at validating the observed results.

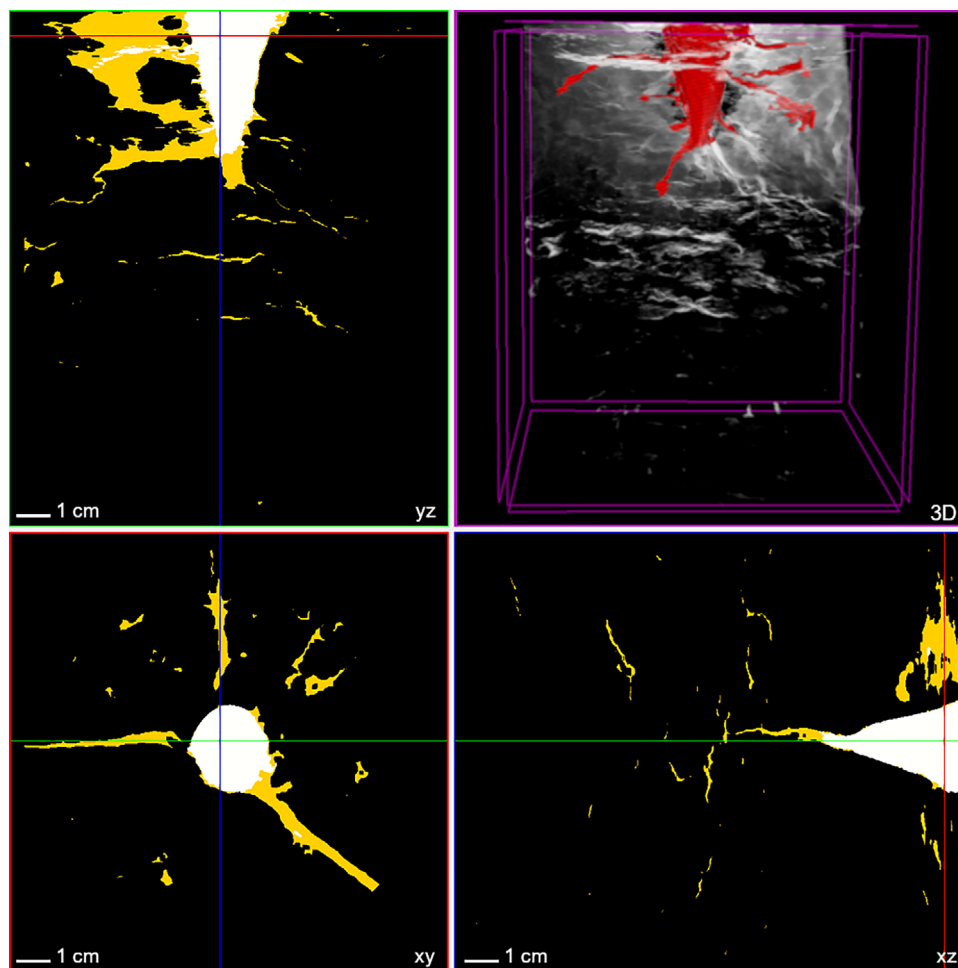


FIGURE 10 Vertical (xz and yz) and horizontal (xy) cross sections from X-ray microtomography (μ CT) imagery of Soil Core 2, and its three-dimensional (3D) perspective representation. The cross sections represent the forage radish (white) and the soil pores (yellow) in two dimensions within the soil matrix (black). The 3D image highlights the whole taproot apparatus (red) surrounded by soil porosity (gray). The other images are reported in the Figures S2–S4.

5 | CONCLUDING REMARKS

The comparison between the soil profiles where FR grew and those on BS highlighted slight differences in terms of soil structure and hydraulic properties. The major change concerned an increase in inter-aggregate porosity and nonequilibrium solute transport due to the action of growing taproots that created fissures and channels along living roots. However, this effect was offset by the deep root system and rapid root extension of FR, which served as an excellent solute scavenger by adsorbing up to 47% of total solute input. Therefore, our starting hypothesis can be partially rejected since solute movement to groundwater was not affected by FR compared to BS conditions. Nevertheless, care must be taken in evaluating the real effectiveness on such cover crop in regulating the solute fate because the experiment was conducted only with living roots that did not generate wider open bio-channels at

the soil surface. Further studies will undoubtedly be needed to investigate water and solute dynamics in the presence of dead roots, for instance, by extending the present study to a full year of FR cultivation.

AUTHOR CONTRIBUTIONS

Carlo Camarotto: Conceptualization; formal analysis; investigation; writing—original draft; writing—review and editing. **Nicola Dal Ferro:** Conceptualization; methodology; writing—original draft; writing—review and editing. **Ilenia Piccoli:** Formal analysis; investigation; writing—review and editing. **Matteo Longo:** Formal analysis; writing—review and editing. **Celine Duwig:** Funding acquisition; supervision; writing—review and editing. **Laurent Oxarango:** Investigation; supervision; writing—review and editing. **Francesco Morari:** Conceptualization; funding acquisition; supervision; writing—review and editing.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

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REFERENCES

- Abdollahi, L., & Munkholm, L. J. (2014). Tillage system and cover crop effects on soil quality: I. Chemical, mechanical and biological properties. *Soil Science Society of America Journal*, 78, 262–270. <https://doi.org/10.2136/sssaj2013.07.0301>
- Abdollahi, L., Munkholm, L. J., & Garbout, A. (2014). Tillage system and cover crop effects on soil quality: II. Pore characteristics. *Soil Science Society of America Journal*, 78, 271–279. <https://doi.org/10.2136/sssaj2013.07.0302>
- Adetunji, A. T., Ncube, B., Mulidzi, R., & Lewu, F. B. (2020). Management impact and benefit of cover crops on soil quality: A review. *Soil and Tillage Research*, 204, 104717. <https://doi.org/10.1016/j.still.2020.104717>
- Allen, R. G., Pereira, L. S., Raes, D., & Smith, M. (1998). *Crop evapotranspiration: Guidelines for computing crop water requirements*. Food and Agriculture Organization of the United Nations.
- Allen, R. G., Pruitt, W. O., Raes, D., Smith, M., & Pereira, L. S. (2005). Estimating evaporation from bare soil and the crop coefficient for the initial period using common soils information. *Journal of Irrigation and Drainage Engineering*, 131, 14–23. [https://doi.org/10.1061/\(ASCE\)0733-9437\(2005\)131:1\(14\)](https://doi.org/10.1061/(ASCE)0733-9437(2005)131:1(14))
- Arganda-Carreras, I., Kaynig, V., Rueden, C., Eliceiri, K. W., Schindelin, J., Cardona, A., & Sebastian Seung, H. (2017). Trainable weka segmentation: A machine learning tool for microscopy pixel classification. *Bioinformatics*, 33(15), 2424–2426. <https://doi.org/10.1093/bioinformatics/btx180>
- Baggs, E. M., Watson, C. A., & Rees, R. M. (2000). The fate of nitrogen from incorporated cover crop and green manure residues. *Nutrient cycling in agroecosystems*, 56, 153–163. <https://doi.org/10.1023/A:1009825606341>
- Ball, B. C., Lang, R. W., Robertson, E. A. G., & Franklin, M. F. (1994). Crop performance and soil conditions on imperfectly drained loams after 20–25 years of conventional tillage or direct drilling. *Soil and Tillage Research*, 31, 97–118. [https://doi.org/10.1016/0167-1987\(94\)90074-4](https://doi.org/10.1016/0167-1987(94)90074-4)
- Batany, S., Peyneau, P.-E., Lassabatère, L., Béchet, B., Faure, P., & Danga, P. (2019). Interplay between molecular diffusion and advection during solute transport in macroporous media. *Vadose Zone Journal*, 18, 180140. <https://doi.org/10.2136/vzj2018.07.0140>
- Beven, K., & Germann, P. (2013). Macropores and water flow in soils revisited. *Water Resources Research*, 49, 3071–3092. <https://doi.org/10.1002/wrcr.20156>
- Blanco-Canqui, H., Mikha, M. M., Presley, D. R., & Claassen, M. M. (2011). Addition of cover crops enhances no-till potential for improving soil physical properties. *Soil Science Society of America Journal*, 75, 1471. <https://doi.org/10.2136/sssaj2010.0430>
- Blanco-Canqui, H., & Ruis, S. J. (2020). Cover crop impacts on soil physical properties: A review. *Soil Science Society of America Journal*, 84, 1527–1576. <https://doi.org/10.1002/saj2.20129>
- Blanco-Canqui, H., Shaver, T. M., Lindquist, J. L., Shapiro, C. A., Elmore, R. W., Francis, C. A., & Hergert, G. W. (2015). Cover crops and ecosystem services: Insights from studies in temperate soils. *Agronomy Journal*, 107, 2449. <https://doi.org/10.2134/agronj15.0086>
- Bodner, G., Leitner, D., & Kaul, H.-P. (2014). Coarse and fine root plants affect pore size distributions differently. *Plant and Soil*, 380, 133–151. <https://doi.org/10.1007/s11104-014-2079-8>
- Camarotto, C., Dal Ferro, N., Piccoli, I., Polese, R., Furlan, L., Chiarini, F., & Morari, F. (2018). Conservation agriculture and cover crop practices to regulate water, carbon and nitrogen cycles in the low-lying Venetian plain. *Catena*, 167, 236–249. <https://doi.org/10.1016/j.catena.2018.05.006>
- Carter, M. R. (1991). Evaluation of shallow tillage for spring cereals on a fine sandy loam. 2. Soil physical, chemical and biological properties. *Soil and Tillage Research*, 21, 37–52. [https://doi.org/10.1016/0167-1987\(91\)90004-H](https://doi.org/10.1016/0167-1987(91)90004-H)
- Çerçioğlu, M., Anderson, S. H., Udawatta, R. P., & Alagele, S. (2019). Effect of cover crop management on soil hydraulic properties. *Geoderma*, 343, 247–253. <https://doi.org/10.1016/j.geoderma.2019.02.027>
- Chen, G., & Weil, R. R. (2010). Penetration of cover crop roots through compacted soils. *Plant and Soil*, 331, 31–43. <https://doi.org/10.1007/s11104-009-0223-7>
- Chen, G., & Weil, R. R. (2011). Root growth and yield of maize as affected by soil compaction and cover crops. *Soil and Tillage Research*, 117, 17–27. <https://doi.org/10.1016/j.still.2011.08.001>
- Chen, G., Weil, R. R., & Hill, R. L. (2014). Effects of compaction and cover crops on soil least limiting water range and air permeability. *Soil and Tillage Research*, 136, 61–69. <https://doi.org/10.1016/j.still.2013.09.004>
- Clark, L. J., Whalley, W. R., & Barraclough, P. B. (2003). How do roots penetrate strong soil? *Plant and Soil*, 255, 93–104. <https://doi.org/10.1023/A:1026140122848>
- Cofer, T. L. (2019). *Radish cover crop growth and compaction alleviation potential in southeastern coastal plain soils* [Doctoral dissertation, Auburn University].
- Cresswell, H., & Kirkegaard, J. (1995). Subsoil amelioration by plant-roots—The process and the evidence. *Soil Research*, 33, 221–239. <https://doi.org/10.1071/SR9950221>
- Dal Ferro, N., Zanin, G., & Borin, M. (2017). Crop yield and energy use in organic and conventional farming: A case study in north-east Italy. *European Journal of Agronomy*, 86, 37–47. <https://doi.org/10.1016/j.eja.2017.03.002>
- Diederik, J., Šimůnek, J., Timmerman, A., & Feyen, J. (2002). Calibration of Richards' and convection–dispersion equations to field-scale water flow and solute transport under rainfall conditions. *Journal of Hydrology*, 259, 15–31.
- Farsad, A., Herbert, S. J., Hashemi, M., & Sadeghpour, A. (2012). An automated suction lysimeter for improved soil water sampling. *Vadose Zone Journal*, 11, vzj2012.0003. <https://doi.org/10.2136/vzj2012.0003>

- Feddes, R. A., Kowalik, P., Kolinska-Malinka, K., & Zaradny, H. (1976). Simulation of field water uptake by plants using a soil water dependent root extraction function. *Journal of Hydrology*, 31, 13–26. [https://doi.org/10.1016/0022-1694\(76\)90017-2](https://doi.org/10.1016/0022-1694(76)90017-2)
- Gabriel, J. L., Quemada, M., Martín-Lammerding, D., & Vanclooster, M. (2019). Assessing the cover crop effect on soil hydraulic properties by inverse modelling in a 10-year field trial. *Agricultural Water Management*, 222, 62–71. <https://doi.org/10.1016/j.agwat.2019.05.034>
- Gao, G., Zhan, H., Feng, S., Fu, B., Ma, Y., & Huang, G. (2010). A new mobile-immobile model for reactive solute transport with scale-dependent dispersion. *Water Resources Research*, 46, 1–16. <https://doi.org/10.1029/2009WR008707>
- García-González, I., Hontoria, C., Gabriel, J. L., Alonso-Ayuso, M., & Quemada, M. (2018). Cover crops to mitigate soil degradation and enhance soil functionality in irrigated land. *Geoderma*, 322, 81–88. <https://doi.org/10.1016/j.geoderma.2018.02.024>
- Giardini, L. (2004). *Productivity and sustainability of different cropping systems*. Patron Editore.
- González, M. G., Ramos, T. B., Carlesso, R., Paredes, P., Petry, M. T., Martins, J. D., Aires, N. P., & Pereira, L. S. (2015). Modelling soil water dynamics of full and deficit drip irrigated maize cultivated under a rain shelter. *Biosystems Engineering*, 132, 1–18. <https://doi.org/10.1016/j.biosystemseng.2015.02.001>
- Gruver, J., Weil, R., White, C., & Lawley, Y. (2014). *Radishes: A new cover crop for organic farming systems*. www.extension.org
- Guo, L., & Lin, H. (2018). Addressing two bottlenecks to advance the understanding of preferential flow in soils. *Advances in Agronomy*, 147, 61–117. <https://doi.org/10.1016/bs.agron.2017.10.002>
- Hallett, P. D., Marin, M., Bending, G. D., George, T. S., Collins, C. D., & Otten, W. (2022). Building soil sustainability from root–soil interface traits. *Trends in Plant Science*, 27(7), 688–698. <https://doi.org/10.1016/j.tplants.2022.01.010>
- Harrigan, T. P., & Mann, R. W. (1984). Characterization of microstructural anisotropy in orthotropic materials using a second rank tensor. *Journal of Materials Science*, 19, 761–767. <https://doi.org/10.1007/BF00540446>
- Haruna, S. I., Anderson, S. H., Nkongolo, N. V., & Zaibon, S. (2018). Soil hydraulic properties: Influence of tillage and cover crops. *Pedosphere*, 28, 430–442. [https://doi.org/10.1016/S1002-0160\(17\)60387-4](https://doi.org/10.1016/S1002-0160(17)60387-4)
- IUSS Working Group WRB. (2015). *World reference base for soil resources 2014, update 2015: International soil classification system for naming soils and creating legends for soil maps* (World Soil Resources Reports No. 106). FAO.
- Jarvis, N. J. (2007). A review of nonequilibrium water flow and solute transport in soil macropores: Principles, controlling factors and consequences for water quality. *European Journal of Soil Science*, 58, 523–546. <https://doi.org/10.1111/j.1365-2389.2007.00915.x>
- Kanzari, S., Ben Nouna, B., Ben Mariem, S., & Rezig, M. (2018). Hydrus-1D model calibration and validation in various field conditions for simulating water flow and salts transport in a semi-arid region of Tunisia. *Sustainable Environment Research*, 28, 350–356. <https://doi.org/10.1016/j.serj.2018.10.001>
- Kristensen, H. L., & Thorup-Kristensen, K. (2014). Root growth and nitrate uptake of three different catch crops in deep soil layers. *Soil Science Society of America Journal*, 68, 529–537. <https://doi.org/10.2136/sssaj2004.5290>
- Ladu, J. L. C., & Zhang, D. R. (2011). Modeling atrazine transport in soil columns with HYDRUS-1D. *Water Science and Engineering*, 4, 258–269.
- Le Bourgeois, O., Bouvier, C., Brunet, P., & Ayrat, P.-A. (2016). Inverse modeling of soil water content to estimate the hydraulic properties of a shallow soil and the associated weathered bedrock. *Journal of Hydrology*, 541, 116–126. <https://doi.org/10.1016/j.jhydrol.2016.01.067>
- Leistra, M., & Boesten, J. J. T. I. (2010). Measurement and computation of movement of bromide ions and carbofuran in ridged humic-sandy soil. *Archives of Environment Contamination and Toxicology*, 59(1), 39–48. <https://doi.org/10.1007/s00244-009-9442-4>
- Levy, B. S., & Chambers, R. M. (1987). Bromide as a conservative tracer for soil-water studies. *Hydrological Processes*, 1, 385–389. <https://doi.org/10.1002/hyp.3360010406>
- Lide, D. R. (2004). *CRC handbook of chemistry and physics* (Vol. 85). CRC Press.
- Linsler, D., Kaiser, M., Andruschkewitsch, R., Piegholdt, C., & Ludwig, B. (2016). Effects of cover crop growth and decomposition on the distribution of aggregate size fractions and soil microbial carbon dynamics. *Soil Use and Management*, 32, 192–199. <https://doi.org/10.1111/sum.12267>
- Longo, M., Delle Vedove, G., Grignani, C., Peressotti, A., Lazzaro, B., Cabrera, M., & Morari, F. (2023). How water table level influences C balance under different fertilization regimes. *Agricultural Water Management*, 289, 108508. <https://doi.org/10.1016/j.agwat.2023.108508>
- Longo, M., Jones, C. D., Izaurre, R. C., Cabrera, M. L., Dal Ferro, N., & Morari, F. (2021). Testing the EPIC Richards submodel for simulating soil water dynamics under different bottom boundary conditions. *Vadose Zone Journal*, 20(5), e20142. <https://doi.org/10.1002/vzj2.20142>
- Marquardt, D. W. (1963). An algorithm for least-squares estimation of nonlinear parameters. *Siam Journal on Applied Mathematics*, 11(2), 431–441. <https://doi.org/10.1137/0111030>
- Marshall, M. W., Williams, P., Nafchi, A. M., Maja, J. M., Payero, J., Mueller, J., & Khalilian, A. (2016). Influence of tillage and deep rooted cool season cover crops on soil properties, pests, and yield responses in cotton. *Open Journal of Soil Science*, 06, 149–158. <https://doi.org/10.4236/ojss.2016.610015>
- Mayer, D. G., & Butler, D. G. (1993). Statistical validation. *Ecological Modelling*, 68, 21–32. [https://doi.org/10.1016/0304-3800\(93\)90105-2](https://doi.org/10.1016/0304-3800(93)90105-2)
- Mencaroni, M., Ferro, N. D., Radcliffe, D. E., & Morari, F. (2021). Preferential solute transport under variably saturated conditions in a silty loam soil: Is the shallow water table a driving factor? *Journal of Hydrology*, 602, 126733. <https://doi.org/10.1016/j.jhydrol.2021.126733>
- Mencaroni, M., Longo, M., Cardinali, A., Lazzaro, B., Zanin, G., Dal Ferro, N., & Morari, F. (2023). Glyphosate and AMPA dynamics during the transition towards conservation agriculture: Drivers under shallow groundwater conditions. *Soil and Tillage Research*, 229, 105659. <https://doi.org/10.1016/j.still.2023.105659>
- Millington, R., & Quirk, J. P. (1961). Permeability of porous solids. *Transactions of the Faraday Society*, 57(8), 1200–1207. <https://doi.org/10.1039/tf9615701200>
- Morari, F. (2006). Drainage flux measurement and errors associated with automatic tension-controlled suction plates. *Soil Science Society of*

- America Journal*, 70, 1860–1871. <https://doi.org/10.2136/sssaj2006.0009>
- Munkholm, L. J., Schjønning, P., Rasmussen, K. J., & Tanderup, K. (2003). Spatial and temporal effects of direct drilling on soil structure in the seedling environment. *Soil and Tillage Research*, 71, 163–173. [https://doi.org/10.1016/S0167-1987\(03\)00062-X](https://doi.org/10.1016/S0167-1987(03)00062-X)
- Nash, J. E., & Sutcliffe, J. V. (1970). River flow forecasting through conceptual models part I—A discussion of principles. *Journal of Hydrology*, 10, 282–290. [https://doi.org/10.1016/0022-1694\(70\)90255-6](https://doi.org/10.1016/0022-1694(70)90255-6)
- Parvin, S., Condon, J., & Rose, T. (2023). Rooting depth and water use of summer cover crops in a semi-arid cropping environment. *European Journal of Agronomy*, 147, 126847. <https://doi.org/10.1016/j.eja.2023.126847>
- Patra, S., Julich, S., Feger, K.-H., Jat, M. L., Jat, H., Sharma, P. C., & Schwärzel, K. (2019). Soil hydraulic response to conservation agriculture under irrigated intensive cereal-based cropping systems in a semiarid climate. *Soil and Tillage Research*, 192, 151–163. <https://doi.org/10.1016/j.still.2019.05.003>
- Piccoli, I., Seehusen, T., Bussell, J., Vitzu, O., Calciu, I., Berti, A., Börjesson, G., Kirchmann, H., Kätterer, T., Sartori, F., Stoate, C., Crotty, F., Panagea, I. S., Alaoui, A., & Bolinder, M. A. (2022). Opportunities for mitigating soil compaction in Europe—Case studies from the soilcare project using soil-improving cropping systems. *Land*, 11, 223. <https://doi.org/10.3390/land11020223>
- Power, M. (1993). The predictive validation of ecological and environmental models. *Ecological Modelling*, 68, 33–50. [https://doi.org/10.1016/0304-3800\(93\)90106-3](https://doi.org/10.1016/0304-3800(93)90106-3)
- Radcliffe, D. E., & Simunek, J. (2018). *Soil physics with HYDRUS*. CRC Press. <https://doi.org/10.1201/9781315275666>
- Ramírez-García, J., Carrillo, J. M., Ruiz, M., Alonso-Ayuso, M., & Quemada, M. (2015). Multicriteria decision analysis applied to cover crop species and cultivars selection. *Field Crops Research*, 175, 106–115. <https://doi.org/10.1016/j.fcr.2015.02.008>
- Rémy, É., & Thiel, É. (2002). Medial axis for chamfer distances: Computing look-up tables and neighbourhoods in 2D or 3D. *Pattern Recognition Letters*, 23(6), 649–661. [https://doi.org/10.1016/S0167-8655\(01\)00141-6](https://doi.org/10.1016/S0167-8655(01)00141-6)
- Ritter, A., Hupet, F., Muñoz-Carpena, R., Lambot, S., & Vanclooster, M. (2003). Using inverse methods for estimating soil hydraulic properties from field data as an alternative to direct methods. *Agricultural Water Management*, 59, 77–96. [https://doi.org/10.1016/S0378-3774\(02\)00160-9](https://doi.org/10.1016/S0378-3774(02)00160-9)
- Sartori, F., Piccoli, I., Polese, R., & Berti, A. (2022). Transition to conservation agriculture: How tillage intensity and covering affect soil physical parameters. *Soil*, 8(1), 213–222. <https://doi.org/10.5194/soil-8-213-2022>
- Scavo, A., Fontanazza, S., Restuccia, A., Pesce, G. R., Abbate, C., & Maurolicale, G. (2022). The role of cover crops in improving soil fertility and plant nutritional status in temperate climates. A review. *Agronomy for Sustainable Development*, 42, 93. <https://doi.org/10.1007/s13593-022-00825-0>
- Schabenberger, O., & Pierce, F. J. (2002). *Contemporary statistical models for the plant and soil sciences*. CRC Press.
- Shipitalo, M., Dick, W., & Edwards, W. (2000). Conservation tillage and macropore factors that affect water movement and the fate of chemicals. *Soil and Tillage Research*, 53, 167–183. [https://doi.org/10.1016/S0167-1987\(99\)00104-X](https://doi.org/10.1016/S0167-1987(99)00104-X)
- Šimůnek, J., & van Genuchten, M. Th. (2008). Modeling nonequilibrium flow and transport processes using HYDRUS. *Vadose Zone Journal*, 7, 782–797. <https://doi.org/10.2136/vzj2007.0074>
- Šimůnek, J., van Genuchten, M. Th., & Šejna, M. (2008). Development and applications of the HYDRUS and STANMOD software packages and related codes. *Vadose Zone Journal*, 7, 587–600. <https://doi.org/10.2136/vzj2007.0077>
- Šimůnek, J., Wendroth, O., & van Genuchten, M. Th. (1999). Estimating unsaturated soil hydraulic properties from laboratory tension disc infiltrometer experiments. *Water Resources Research*, 35, 2965–2979. <https://doi.org/10.1029/1999WR900179>
- Sundermeier, A. (2008). *Oilseed radish cover crop* (Extension Fact Sheet SAG-5-08). Ohio State University.
- Thorup-Kristensen, K., Magid, J., & Jensen, L. S. (2003). Catch crops and green manures as biological tools in nitrogen management in temperate zones. *Advances in Agronomy*, 79, 227–302. [https://doi.org/10.1016/S0065-2113\(02\)79005-6](https://doi.org/10.1016/S0065-2113(02)79005-6)
- Tonitto, C., Li, C., Seidel, R., & Drinkwater, L. (2010). Application of the DNDC model to the Rodale Institute Farming Systems Trial: Challenges for the validation of drainage and nitrate leaching in agroecosystem models. *Nutrient Cycling in Agroecosystems*, 87, 483–494. <https://doi.org/10.1007/s10705-010-9354-8>
- van Genuchten, M. Th. (1980). A closed-form equation for predicting the hydraulic conductivity of unsaturated soils. *Soil Science Society of America Journal*, 44, 892–898. <https://doi.org/10.2136/sssaj1980.03615995004400050002x>
- van Genuchten, M. Th., & Wierenga, P. J. (1976). Mass transfer studies in sorbing porous media. I. Analytical solutions. *Soil Science Society of America Journal*, 40, 473–480. <https://doi.org/10.2136/sssaj1976.03615995004000040011x>
- Vervoort, R. W., Radcliffe, D. E., & West, L. T. (1999). Soil structure development and preferential solute flow. *Water Resources Research*, 35, 913–928.
- Villamil, M. B., Bollero, G. A., Darmody, R. G., Simmons, F. W., & Bullock, D. G. (2006). No-till corn/soybean systems including winter cover crops. *Soil Science Society of America Journal*, 70, 1936–1944. <https://doi.org/10.2136/sssaj2005.0350>
- Vogel, H.-J., Weller, U., & Schlüter, S. (2010). Quantification of soil structure based on Minkowski functions. *Computers & Geosciences*, 36(10), 1236–1245. <https://doi.org/10.1016/j.cageo.2010.03.007>
- Wahlström, E. M., Hansen, E. M., Mandel, A., Garbout, A., Kristensen, H. L., & Munkholm, L. J. (2015). Root development of fodder radish and winter wheat before winter in relation to uptake of nitrogen. *European Journal of Agronomy*, 71, 1–9. <https://doi.org/10.1016/j.eja.2015.07.002>
- Weil, R., White, C., & Lawley, Y. (2009). *Forage radish: New multi-purpose cover crop for the Mid-Atlantic* (Vol. 824) [Fact sheet]. University of Maryland Cooperative Extension. http://covercrop.org/wp-content/uploads/2016/05/Weil_radish.pdf
- White, C. M., & Weil, R. R. (2011). Forage radish cover crops increase soil test phosphorus surrounding radish taproot holes. *Soil Science Society of America Journal*, 75(1), 121–130. <https://doi.org/10.2136/sssaj2010.0095>
- Williams, S. M., & Weil, R. R. (2004). Crop cover root channels may alleviate soil compaction effects on soybean crop. *Soil Science Society of America Journal*, 68, 1403–1409. <https://doi.org/10.2136/sssaj2004.1403>

Yunusa, I. A. M., & Newton, P. J. (2003). Plants for amelioration of subsoil constraints and hydrological control: The primer-plant concept. *Plant and Soil*, 257, 261–281. <https://doi.org/10.1023/A:1027381329549>

SUPPORTING INFORMATION

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